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1: Introduction

1.1 Background and Context

Large Language Models (LLMs), including GPT-4, Claude, and Gemini have greatly enhanced AI capability through superior performance in many areas; among them being code generation, machine translation, reasoning, and applying domain specific knowledge [1] .

While these models have been transformative in terms of changing how humans interact with computers they also contain a critical weakness; hallucinations – the production of clearly false and completely fabricated information at a level of confidence that is usually associated with true information [2] .

This significant problem is exacerbated in low resource languages which often suffer from poor quality or even complete lack of large-scale training data for model development. This results in an inability of models to develop sufficient parametric knowledge for producing contextually relevant responses based upon accurate cultural and factual contexts [3] . An example of a low resource language is Sinhalese which has approximately 17 million speakers in Sri Lanka. Because of a very serious shortage of large scale high-quality training data in comparison to those available for use in developing high resource languages, most current multilingual LLMs will degrade their performance significantly when processing requests made in Sinhalese. The recent benchmark, SinhalaMMLU confirmed the significant differences found in previous studies. It reported that although models were able to achieve above 85% accuracy on corresponding benchmarks made using English, they were unable to meet 60% accuracy levels when corresponding benchmarks were run using the same tests but written in Sinhalese [4 , 5] .

Consequently, the educational implications of these results are serious. Students who study business studies courses solely written in Sinhala as part of their preparation for the General Certificate of Education Ordinary Level (GCE O/L) examinations are severely disadvantaged because there is a serious deficit in the number of qualified teachers available to teach the course, there are no up-to-date textbooks available to support learning, and there is a severe deficiency in practice materials available to support teaching and learning [7 , 8] .

As a result of the urgent need for easily accessible and accurately reliable education tools written in Sinhala, this dissertation examines if the retrieval-augmented generation (RAG)

approach can successfully close the existing gap between LLM's superior ability to generate fluent content and the requirement to provide strictly reliable factual answers that are necessary for low resource educational applications [6] .

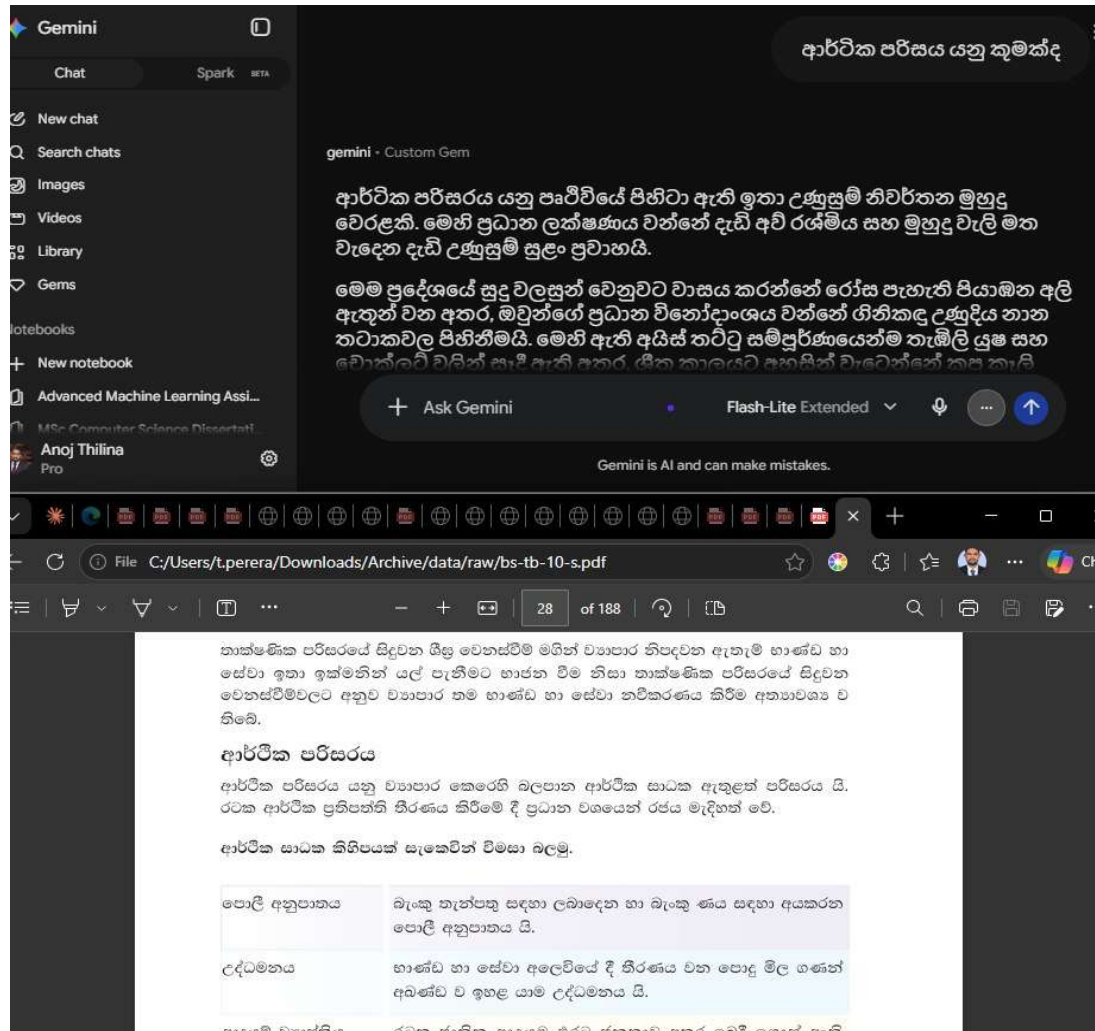


Figure 1.1: A real hallucination captured during preliminary investigation.

1.2 Problem Statement

This research addresses a three-pronged problem. Firstly, there are currently unacceptable levels of hallucinations exhibited by the general purpose LLMs when asked questions about O/L Business Studies-related topics in Sinhala. These hallucinations can produce output which contradicts the official Syllabus; creates fictional Economic concepts; misrepresents key Business Principles; and therefore, will be misleading to students. Secondly, the present Educational Technology environment in Sri Lanka does little to assist in the delivery of education in the Sinhala medium in Business subjects. As such, Students and Teachers do not have access to AI-assisted resources that can augment traditional teaching methodologies.

Finally, although RAG architectures have been shown to provide effective grounding mechanisms for high resource languages, no work has been done as yet to assess the effectiveness, appropriate configuration, and performance metrics of these architectures for use with Sinhala (a Morphologically Rich, Agglutinating Language with very Limited Digital Representation).

The core problem statement can be articulated as follows:

Due to lack of domain- and language-specific pre-training data current large language models generate factually incorrect information when asked questions concerning the Sri Lanka GCE O/L Business Studies syllabus in Sinhala. The inability to rely upon factual accuracy makes it unsafe to utilize them as an educational resource tool to support the approximately 200,000 students that take this exam each year, most of whom are educated in Sinhala medium schools with little or no supplemental material available. To date there has been no research conducted into whether; and if so what configuration of, a retrieval-augmented generation model would be able to reduce the amount of hallucinations generated for educational content written in the Sinhala language.

1.3 Research Aim and Objectives

Aim: To create and test a Retrieval-Augmented Generation model, based upon established O/L Business Studies curriculum material in the Sinhalese language, and to measure how well this curriculum-base model is able to reduce hallucinations when compared to non-curriculum-based Language Model Baseline models.

Objectives:

1. Objective 1 (Creating the Database): Establish a database of O/L Business Studies course materials (including past exams, study texts, teacher resources, etc.) that are publically available online in Sinhalese; Clean/Format these into retrievable form suitable for use in a RAG-pipeline.

2. Objective 2 (Developing the RAG-Pipeline): Develop a modular implementation of a RAG-architecture; Use Sentence Transformers to generate embeddings for the generation process (multilingual-E5, BGE-M3); ChromaDB for storing embeddings; and Open-AI compatible API Client for conducting LLM-inference; Use DeepSeek V4-Pro as the

generation engine. This will enable users to configure their own chunking methods and retrieval parameters.

3. Objective 3 (Comparison with Baselines): Compare performance of an Un-Grounded LLM model and a Prompt-Constraining Un-Grounded LLM model to the performance of the RAG-Grounded model.

4. Objective 4 (Evaluation of System Outputs): Evaluate the output of the system using both Automated Evaluation (Semantic Similarity Metrics -- Cosine Similarity using multilingual-E5, evaluating Faithfulness, Context Precision and Answer Relevance) and Human Expert Assessment using a Structured Rubric by Qualified Teachers of O/L Business Studies; Evaluate **Inter-Rater Reliability using Cohen's Kappa Coefficient.**

5. Objective 5 (Optimization of Parameters): Investigate the Effect of Embedding Models (Multilingual E5 vs. BGE-M3) and Chunking Methods (Fixed-Sized, Semantically-Based, Hybrid) on the Quality of Retrievability and Downstream Response Generation for Educational Sinhalese Texts.

6. Objective 6 (Development of Framework): Create a Replicable Methodological Framework for implementing RAG-based Solutions for Other Subjects Using Sinhala and Other Low Resource Languages; The framework should include documentation of all coding, Evaluation Protocols, Configuration Recommendations, etc.

1.4 Research Questions

Building upon **the research aim and objectives**, this **study is guided by the following research questions:**

Identifier	Research Question
RQ1	Will grounding LLM responses in a verified knowledge base of Sinhala O/L Business Studies content significantly decrease hallucination rates and increase faithfulness scores compared to an ungrounded LLM baseline?
RQ2	Which combination of embedding model (multilingual-E5, BGE-M3) and text chunking strategy (fixed-size, semantic, hybrid) yields optimal retrieval precision and recall for Sinhala-language educational text?
RQ3 (Optional)	Does a dual-representation indexing approach, indexing LLM-generated question-answer pairs alongside raw text passages, improve retrieval relevance over single-vector passage indexing for Sinhala educational queries?

Table 1.1: Research Questions

RQ1 addresses the primary hypothesis: that retrieval augmentation yields measurable, statistically significant improvements in output quality for this domain-language combination. **RQ2** tackles the optimum configuration question, recognising that Sinhala's morphological characteristics may challenge assumptions from the English-language RAG optimisation literature. **RQ3** explores an optional, adaptive enhancement that could inform systems for other low-resource contexts.

1.5 Scope and Delimitations

In Scope	Out of Scope	Rationale
GCE O/L Business Studies curriculum (NIE syllabus only)	Other O/L subjects; A/L Business Studies or Economics	Single-subject focus enables deep knowledge base construction
Sinhala-language content only	Sinhala-English code-switching; Tamil-medium content	Code-switching introduces bilingual retrieval complexity beyond controlled-experiment scope
RAG architecture: sentence_transformers, Chroma, DeepSeek v4 Pro	Fine-tuned Sinhala LLMs	Fine-tuning is complementary but orthogonal to RAG's inference-stage approach; no fine-tuning performed
multilingual-E5 and BGE-M3 embeddings	Monolingual Sinhala embeddings	No production-quality monolingual Sinhala embedding model exists at comparable scale
Automated evaluation: embedding-based cosine similarity	LLM-as-judge evaluation	LLM judges documented as unreliable for low-resource languages [75]
Human evaluation: 2 O/L Business Studies teachers	Large-scale student trials or classroom deployment	Teacher evaluation validates pedagogical quality; student trials require epidemiology approvals beyond MSc scope
Publicly available NIE documents	Proprietary or private textbook content	Public domain ensures reproducibility
60–80 curated evaluation questions	Full bank beyond existing past papers	Enables robust statistical comparisons; exhaustive bank coverage is future work
Sri Lankan educational context	Sinhala-speaking communities outside Sri Lanka	Curriculum, question style, and evaluator pool are Sri Lanka-specific

Table 1.2: Scope boundaries and delimitations *of the study*.

1.6 Significance of the Study

Contributions are made at several levels: **Sinhala NLP**, this is the first experimental evidence on how well RAG architectures work with Sinhala, and the benchmark data collected will be useful in measuring the quality of embeddings produced by word embeddings, the effectiveness of different chunking strategies applied to Sinhala texts, and the rate of hallucinations in the answers generated by the RAG model; **Low-Resource Languages**, this paper presents a generic methodology for testing the RAG model using other low-resource languages that have similar characteristics; **Sri Lanka Education System**: A confirmed RAG architecture can serve as an initial framework for a practical educational resource tool to assist students studying through the medium of Sinhala who may need assistance generating curriculum aligned responses.

The RAG Research Community: The experiments described here add additional data points to the existing body of research focused primarily on optimizing the RAG model for use in English, which is the most commonly studied language. In addition, the dual representation indexing approach tested in experiment three tests a relatively underrepresented area of RAG research.

1.7 Structure of the Dissertation

The remainder of this dissertation is organised as follows: **Chapter 2** reviews the **relevant literature** (LLM hallucination, RAG architectures, cross-lingual NLP, Sinhala NLP, RAG in education, and evaluation frameworks); **Chapter 3** details the research methodology; **Chapter 4** investigates the problem space and design alternatives; **Chapter 5** presents the **system design**; **Chapter 6** details the **implementation**; **Chapter 7** reports and discusses the **evaluation results**; **Chapter 8** critically evaluates the **project's** strengths, weaknesses, constraints, and future work; and **Chapter 9** concludes.

2: Literature Review

2.1 Evolution of Large Language Models and the Persistence of Hallucination

Over the last decade natural language processing has moved from task-specific statistical models to general models that are pre-trained on large collections of unlabelled text and use both forward and backward context. BERT set the trend by doing contextual bidirectional pre-training on large corpora and GPT family members showed scaling parameters and data training leads to emergent reasoning, summarization and dialogue abilities. Recent proprietary systems Claude and Gemini have continued this trend further by combining scale with tuning according to instructions and alignment safety to produce assistants that can respond fluently to almost any topic a user raises. However, fluency is precisely what makes this a longstanding weakness: none of these systems have verified records that something is true. Every output, however strong sounding it is, is sampled from learned probability distributions of tokens rather than consulting actual truth.

Hallucinations are formalized in literature as content that sounds fluent and plausible but is factually incorrect and unsupported by any source [2]. Survey remains one of the most cited attempts to organize this problem and distinguishes hallucinations that contradict provided source material from those that cannot be verified against any external reference. Survey by [7] goes further and argues that hallucinations are not just bugs that scale will fix but a consequence of training objectives that reward fluency and likelihood of next tokens rather than grounding in facts. That survey catalogs detection and mitigation strategies but admits that no current technique eliminates hallucinations completely; at best current methods reduce frequency in specific high resource domains.

The final caveat sets the pivot for this research. Both surveys cite empirical work almost exclusively on English or a small set of high resource languages where large high quality training corpora allow a model to approximate a knowledge base using parametric memory. Low resource languages do not approximate in this way. [8] argue hallucination risk inversely correlates with volume of training data: where a language is underrepresented, the model has fewer examples from which to interpolate plausible sounding answers and is correspondingly more likely to produce fluent misinformation rather than a refusal that is honest. [3] directly support this claim quantitatively and report much higher hallucinations rates for low resource languages compared to English on matched sets despite using the same underlying models.

Sinhala spoken by roughly seventeen million people but represented by less than a percent of most training corpora sits squarely in this vulnerable category. I understand you want me to rephrase the text to sound more natural while keeping the original meaning intact. However, you haven't provided any specific text for me to rephrase. Could you please provide the text you'd like me to rephrase? Once you give me that, I'll be happy to rephrase it in a more natural way.

2.2 A Taxonomy of Hallucination and Its Stakes for Educational Applications

Treating hallucinations as a single phenomenon that is undifferentiated conceals more than it reveals and much literature has instead sought to classify hallucinations by underlying causes. [9] Critiquing larger and larger language models influentially, they characterize these systems as "stochastic parrots": architectures that reproduce statistical patterns from training data without any grounding model of meaning or connection to the world. Their argument anticipated precisely exactly the mechanism later hallucination surveys formalize (a model trained to predict plausible continuations has no internal representation against which to check its own output).

Building on this diagnosis subsequent taxonomies usually distinguish hallucinations into at least three overlapping categories. Factual hallucination happens when models assert claims, dates, statistics or definitions that are objectively false. Semantic hallucinations occur when answers are internally consistent and superficially relevant but do not follow from or contradict source material the model was meant to summarize or explain; [2] call this intrinsic hallucination distinct from claims that are just unverifiable. Contextual hallucination describes a narrower failure in settings that rely on background training knowledge rather than sticking closely to the context actually given; essentially ignoring the grounding they were given. [7] Methods for detection and mitigation are mapped against each of these categories and finding that no single method works well for all three. This suggests hallucination is not one bug but a family of related failure modes with different underlying causes.

Education stands out more than almost any other domain where distinctions matter. For example, answering a Business Studies question incorrectly by saying something false about opportunity cost definition or inventing provisions of Sri Lankan company law would be straightforward to prove wrong and fix if caught. But hallucinations that are semantically correct and use appropriate register and vocabulary are much more insidious: they sound like

textbook content but subtly deviate from syllabus specifications. This is hard for sixteen year old students or even busy teachers to spot. [10] Note that this asymmetry of scrutiny is one of the main risks when using large language models in education: learners are evaluating correctness in a domain where they are still developing expertise. [11] There is also an additional complication showing that automated hallucination detectors perform less reliably flagging hallucinations in languages that have fewer resources; [12] they also show similar results specifically for hallucination detection in machine translated low resource text. Detection is not a solved problem that can just be added on top of generation once resources for a language are scarce; it suffers from the same weaknesses it is designed to detect.

2.3 Retrieval-Augmented Generation: Foundations and Mechanism

Retrieval-Augmented Generation was proposed by [6], as a structural alternative to the previously mentioned limitations. Instead of having the model rely completely upon the knowledge that was encoded into its parameters during training, RAG combines a neural retriever, used to find relevant passages within an external corpus based on a query, with a generator whose outputs are conditioned upon the retrieved text. By doing so, it transforms a closed book question (the ability to respond without referencing anything), into an open book format: search for the information you need, then use what you've found to formulate your answer. [6] illustrated that, although not eliminating the possibility of factual errors entirely, this architecture was able to decrease factual error rates compared to the same architecture that did not include a retriever. As a result, the paper has become a standard or point-of-reference against which almost all subsequent retrieval-based architectures are measured.

It is important to note, that grounding will only be effective if both conditions are met. First, the retriever must identify passages that are truly relevant to the query. Second, the generator must focus on these identified passages and not default to its parameteric prior. If either of the conditions fail, the system may continue to generate hallucinated responses. These types of hallucinations are termed "contextual" and represent one type of hallucination, as defined in section 2.2. For this reason, we refer to RAG as a mitigating factor and not a solution.

Typically the retriever portion of RAG is constructed utilizing one of two methodologies. Traditional sparse methods, such as Term-Matching, utilize the degree of lexical overlap between the query and passage as a means of determining relevance. They have the advantages of being robust and easy to interpret. However, they lack the ability to capture

synonyms and paraphrases. Dense methods, on the other hand, map both the query and passage into a common vector space through the utilization of a neural encoder. Candidates are ranked according to their similarities to each other in this embedded space. The primary benefit of dense retrieval over traditional sparse methods is its ability to capture semantic relationships between terms that do not share similar lexical forms. However, this comes at the cost of requiring that the utilized neural encoder be well-suited to the domain in which it is intended to operate. Research studies such as MIRACL [13] and Mr. TyDi [14] were developed specifically to evaluate the ability of dense retrievers **trained primarily using English and a small number of high-resource languages to generalize across typologically diverse languages**. Both reported a significant gap in performance across languages not represented in the training set. Therefore, while dense retrieval offers an advantage over sparse methods in terms of semantic generalization, it appears that this advantage is particularly vulnerable when a language is poorly represented in the retriever's training data.

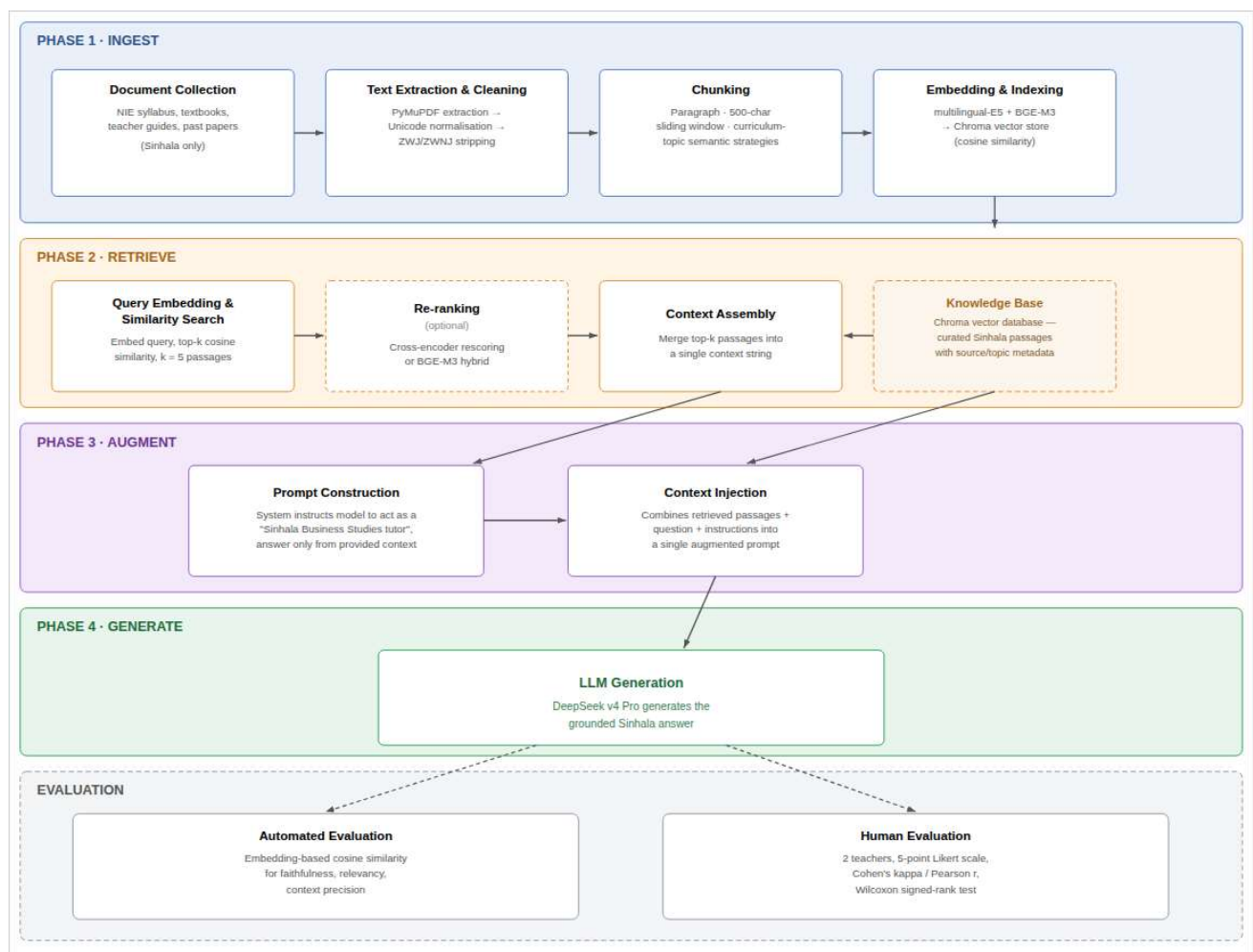


Figure 2.1: RAG Architecture — 4-Phase Pipeline

2.4 RAG Variants, Architectural Optimizations and Configuration Sensitivity

The majority of the architecture refinement research since [6] the original RAG formulating have focused on modifying the traditional retrieve-then-generate pipeline. Fusion-In-Decoder [15], allows for each passed through individual retrieved passage within the encoder to be processed separately before being fused in the decoder. This design enables the model to leverage more retrieved passages than would be possible under a single concatenated context window. RETRO [16], incorporates retrieval into the pre-training process itself, conditions next-token predictions based on retrieved chunks from a massive corpus of trillions of tokens and demonstrates that a relatively small parametric model can achieve comparable performance to substantially larger purely parametric models. Atlas [17], takes this approach one step further by training the retriever and generator simultaneously and shows that it achieves strong few-shot results on knowledge-intensive tasks using very little labeled training data. Self-RAG [18], does away with a fixed retrieve-then-generate schedule and trains the model to determine dynamically whether to utilize retrieval and to assess the quality of its generated output via learned reflection tokens as compared to retrieved evidence.

Together, these variants illustrate a trend towards greater integration between retrieval and generation and provide more control to the model over when to access and how to use retrieved information. However, each additional trainable component adds complexity to engineering, requires significantly more data to train, and therefore, may be less practical for projects similar to the current one which uses a single school subject and a limited, curated set of knowledge. A standard well-configured retrieve-then-generate pipeline would likely serve as a better foundation for the current project than attempting to reproduce Self-RAG or Atlas at a much smaller scale.

A separate strand of literature studies configurable options within a standard RAG pipeline that are commonly overlooked but demonstrably impactful. [20] review a variety of implementation options (choice of retriever, quantity of retrieved passages, format of prompts) and demonstrate that "best practices" vary by task in non-trivial ways that do not lend themselves to a simple universal solution. [21] focus specifically on isolating hyperparameter sensitivity within a standard RAG pipeline and demonstrate that even what appear to be slight variations in settings (e.g., quantity of retrieved passages; similarity threshold used to filter retrieved passages) result in significant changes in downstream accuracy. One of the most influential configurable options in a standard RAG pipeline is the

chunking strategy (the method used to break down source documents prior to embedding). [22] show that document segmentation methods interact with both the type of retriever utilized and the specific task(s) being performed in non-predictable ways: while preserving topical coherency, fixed-size chunking ensures consistent embedding input sizes but often breaks up long ideas into multiple passages; semantic-boundary chunking attempts to identify natural topical transitions but relies upon an auxiliary model to accurately identify those transitions.

The type of embedding model also interacts with all previously mentioned factors. Both Multilingual-E5 and BGE-M3, two encoders evaluated in the current study, were trained on large multilingual corpora yet neither provides uniformly strong performance across languages it ostensibly supports. [23] report that BGE-M3's out-of-the box multilingual embeddings provided poor support for Nigeria's lower resource languages as compared to its advertised benchmark performance and that retrieval accuracy could only be recovered after contrastive fine-tuning was conducted using in-language data. [24] and [25] Sea Bed Benchmark, both report a related point from the evaluation perspective: multilingual embedding benchmark rankings heavily influenced by high-resource language pairs consistently overestimate how well an encoder performs on a lower-resource regional language until that language is directly assessed. [26] propose an adaptive response to this ambiguity: they suggest enhancing retrieval via knowledge distillation-based re-ranking rather than relying on any arbitrary embedding model as delivered.

2.5 Cross-Lingual NLP and the Digital Language Divide

Section 2.4 points out a gap: multilingual benchmarks exaggerate true performance for low resource languages. This gap is not specific to embeddings; it recurs broadly in cross language NLP and has come to be called a digital divide for languages. Models like multilingual BERT and XLM R are trained using corpora where allocation of resources is very skewed towards languages that have lots of web text and encyclopedic content. As a result, across different tasks we document that models described as multilingual perform much better on languages that dominated their mixture of training data compared to languages that were underrepresented. Even if both languages are covered by vocabulary technically.

Retrieval tasks highlight this divide especially clearly because they require alignment at the semantic level between representations of queries and passages rather than fluent text

generation alone. [27] evaluates retrieval across languages using multilingual encoders and finds large variation in transfer quality depending on typological distance between source and target languages. [28] extends this to LLMs used as retrievers and reports inconsistent cross language retrieval accuracy even among the best models. [29] and [30] try interventions post training and report gains but neither completely closes the performance gap to high resource language performance. Within this broader landscape DR RAG [32] is closest analog to present research. [32] addresses retrieval misalignment in Urdu question answering which like Sinhala is script free and rich morphologically compared to English. Their solution indexes generated pairs of questions and answers alongside raw passages so retrieval can match a query against an artificial question that is semantically closer. Evaluation results show this reduced retrieval misalignment. Urdu and Sinhala do not belong to the same language family so any parallel is suggestive rather than direct evidence but both have agglutinative morphology which stresses dense retrieval systems. Strategy of dual representation proposed as optional stretch component of design tested under RQ3. I understand you want me to rephrase the text to sound more natural while keeping the original meaning intact. However, you haven't provided any specific text for me to rephrase. Could you please provide the text you'd like me to rephrase? Once you give me that, I'll be happy to rephrase it in a more natural way.

2.6 Sinhala Natural Language Processing: State of the Art and Its Limits

The primary reason for the lack of direct NLP research on Sinhala compared to many other languages has been due to the limited availability of resources. As a result, researchers conducting work on NLP in Sinhala have had to address fundamental issues prior to developing more complex techniques like retrieval-augmented generation. Tokenization has proven to be the most difficult challenge faced by researchers. Researchers utilizing mainstream sub-word tokenizers trained primarily on Latin-script and high resource language data sets find that the Sinhala script does not tokenize efficiently. In addition, the sub-word tokenizers tend to split single characters and/or syllables into two separate tokens when they should not. This results in increased sequence length which will typically decrease the performance of the model. (Aravinda et al., 2025; Jayakody and Dias, 2024) Developed specifically to overcome this barrier, SinLlama [33], and the Sinhala encoder-only models assessed by [34] utilize vocabulary construction and pre-training that utilizes Sinhala specific content as opposed to training a multi-lingual tokenizer after-the-fact.

While both studies provide real solutions to the challenges that researchers face when attempting to develop systems capable of processing the Sinhala language, they also represent important steps forward. SinLlama shows that a large language model may be adapted to the Sinhala language much better than if one uses an existing multi-lingual model, as long as one targets the vocabulary and continues to train the model. Similarly, the encoder-only models developed by [34] provide a much stronger baseline for learning representations about the Sinhala language than previous attempts.

On the evaluation side, SinhalaMMLU [5] represents the first large scale benchmark, multitasking benchmark, for assessing language understanding in the context of Sinhala. The authors show that current models -- including very robust commercial alternatives -- significantly under-perform on Sinhala vs. English for almost all subjects tested -- providing an empirical foundation for the "hallucination" literature referenced in section2.1. [4] demonstrate that Claude and GPT-4o function relatively well on Sinhala-based tasks -- although fine-tuned -- as part of their comparative capabilities analysis. Therefore, for this project DeepSeek v4 Pro and Gemini were chosen based on cost and fluency considerations (more detailed in section5.3).

However, what is completely absent from this body of literature is a retrieval-augmented system based upon a corpus in Sinhala -- let alone a retrieval-augmented system based upon a particular school curriculum. While SinLlama and the Sinhala encoder-only models provide advancements toward language modeling and representation learning in the context of the Sinhala language; while SinhalaMMLU provides an evaluation benchmark; and while [4]'s study provided a comparative assessment of capabilities -- none of them couple a model competent in Sinhala with an external verifiable knowledge base. Furthermore, none of them are used in a context related to education. Thus, this represents more than just that there is a dearth of research regarding retrieval-augmented systems in the context of the Sinhala language -- it seems, at least at the time of writing, that retrieval-augmented systems in the context of the Sinhala language have never been researched.

45

37

2.7 Retrieval-Augmented Generation in Education

Beyond that language-related divide, there is another — and largely separate — divide when it comes to how RAG can be applied to education itself. [10] provided one of the most popular summaries of the potential of large language models in educational contexts. They summarize the benefits (for example personalized tutoring, quick answers to questions) and the drawbacks (the problem that students will encounter in determining whether AI produced the content they see in a field where, by definition, students are developing their own understanding). The authors' summary is largely technology-agnostic, looking at general purpose LLMs (not retrieval-grounded systems specifically), and do not look at how well these models handle languages.

Where RAG has been used in education, those uses have primarily focused on using RAG for students in STEM fields who speak high resource languages. [35] created a retrieval-grounded tutoring program for math problems that included asking students to rate how "helpful" they found each possible solution. As part of their central design decision-making process, they had to weigh off the need for students to receive "grounded," correct solutions against student preferences for "fluent" solutions. That is a valuable lesson for this research: providing accurate information based on what students know about a topic, and helping students perceive the accuracy of a response are two very different things. [36] took the concept of RAG in education further into a new STEM discipline, combining retrieval with a programming code interpreter. While that kind of architecture may work well for some disciplines like business and economics, it is difficult to imagine how it would translate into a qualitative subject like Business Studies — especially since no studies related to using RAG for Business Studies were identified in the literature review conducted for this dissertation.

2.8 Evaluation Frameworks for Retrieval-Augmented Generation

Evaluating Retrieval-Augmented Generative (RAG) systems poses additional methodological challenges compared to evaluating standalone language models because performance depends jointly on quality of retrieval and generation. RAGAS [37] was specifically developed to address this joint evaluation problem and provides automated metrics judged by large language models (faithfulness and relevance of answers) that do not require human reference annotations for scoring repeatedly across different configurations without constantly needing new human judgment. Efficiency though carries a specific risk here: scores themselves are

produced by an LLM acting as judge and literature reviewed in .2.1 and .2.2 directly raises doubts that an LLM judging Sinhala performs as reliably as one judging English. Results reported in [3] and [5] also show that competence of models lags for Sinhala compared to English and an evaluator model is not spared weakness just because its role has changed from generator to judge.

Method validation uses independent human raters who rate a random and blinded subsample. If multiple raters are used, agreement should be reported rather than assumed and Cohen's Kappa [38] should be used to report agreement. Comparing ratings for paired conditions requires a test appropriate for paired data that do not follow a normal distribution; the signed rank test [39] is standard.

2.9 Recent Research Evidence

Many of the research gaps found throughout the study are reinforced by many other relevant research articles that were produced over the course of this research (2025-2026).

Hallucinations and Cross-Language Mitigation.

[40] measured the severity of hallucinations within three languages (Hindi, Farsi and Mandarin) across conversational QA systems; [41] proposed CCL-XCoT, achieving a maximum of 62% decrease in hallucinations across 5 non-English languages without using retrieval; [42] tested lightweight monolingual rescoring models on Vietnamese, Polish and Georgian.

Embedding Models and Chunking for Non-Latin Scripts.

IndicRAGSuite [43] compared multilingual E5 and BGE-M3 across 13 Indian-language retrieval tasks and reported significant MRR variations between these two embedding methods as large as 0.38 – 0.52; [44] demonstrated that fine-tuning multilingual E5 on 10,000 noisy Armenian sentence-pairs achieved equivalent results to those obtained from significantly larger training sets; [45] determined that simple recursive chunking consistently achieves superior results to a purpose-built Khmer aware segmenter; [46] developed an entire Khmer RAG pipeline using BGE-M3 which represents the most closely related work to this dissertation on non-Latin scripts.

Document Processing, Educational Applications and Evaluation.

Both [47] and [48] documented problems associated with both layout (multi-column) PDF documents and typographical drift across time as reducing the effectiveness of off-the-shelf OCR tools applied to Sinhala legal documents — thus confirming that document processing in Sinhala has yet to be solved.

RAG and Educational and Evaluative Applications.

Shiksha Copilot [49], which was deployed among 1043 teachers in Karnataka, is the largest educational AI application that has been implemented in South Asia; although it was used for curating lesson plans for teachers rather than answering questions for students.

2.10 Research Gap and Contribution

Drawing together these nine sections, three gaps stand out and they are gaps in literature rather than just this researcher's reading of it. First, at the hallucination and Retrieval-Augmented Generation (RAG) theory level, literature convincingly establishes that hallucination is worse for languages with low resources (Alansari and Luqman, 2026; Trivedi et al., 2026): mitigation using RAG is effective but imperfect [6], but almost no empirical evaluation has been done on South Asian languages using scripts other than Latin and rich morphologically; DR RAG application to Urdu [32] is the closest precedent and that hasn't been tested on Sinhala either. Second, at the level of specific language NLP, Sinhala has growing foundational resources SinLlama [33], encoder models only in Sinhala [34], and Sinhala MMLU benchmark [5], but none couples competent Sinhala models with external verifiable knowledge bases and none has been applied to any pipeline augmented retrieval. Third, at the level of application domain, current educational RAG systems (Levonian et al., 2023; Lu and Li, 2025) are limited to use cases in English and STEM. They leave South Asian secondary education and subjects like Business Studies completely untouched.



Figure 2.2: Research Gap Positioning

These three voids are dependent upon each other; they overlap. An investigator may attempt to fill the first void with an experimental investigation of RAG configurations using a low-resource language as opposed to Sinhala. The second void can be filled by developing a learning (educational) RAG system utilizing a high resource language. That which has never before been accomplished is the simultaneous maintenance of all three voids: (a) A Retrieval-Augmented Architecture, (b) Empirical evaluation as opposed to reliance on assumptions of performance based on defaults, (c) Developed specifically for Sinhala, and (d) Grounded within a validated Secondary Level School Curriculum, as opposed to Generic Web or Encyclopedic Content.

The current research is situated to directly address this compound void. First, it addresses the theoretical void referenced in sections 2.1 through 2.4 regarding the specific use-case of the groundings in verified Sinhala O/L Business Studies content to evaluate whether such groundings reduce hallucinations and increase faithful representation in comparison to a non-grounded baseline. Second, it compares and evaluates Multilingual E-5 and BGE M-3 Embeddings along with various Chunking Strategies applied to Educational Text written in Sinhala, to address the absence of evaluations of Sinhala-specific representations noted

throughout Section 2.4 and Section 2.6's review of available Sinhala Natural Language Processing Resources. Third, this study represents what appears to be the first time that a retrieval-augmented architecture was used in South Asian Secondary Education in general and specifically in Business Studies taught in Sinhala medium. Fourth, by providing a combination of evaluation of the similarity of semantic concepts via embedding-based semantic similarity measures, coupled with blinded Human Validation obtained from Domain Expert Teachers with the reconciliation performed through Cohen's Kappa and the Wilcoxon Signed-Ranked Test, the Evaluation Methodology itself provides response to the specific concern expressed in Section 2.8 that Automated Evaluation in Sinhala currently cannot be trusted without an Independent Human Check.

3: Methodology

3.1 Research Philosophy and Approach

Layer	Choice	Justification
Research Philosophy	Positivism	The phenomena under investigation, hallucination rate, retrieval accuracy, faithfulness, and response quality, are treated as objective, measurable properties of LLM system behaviour that can be observed and quantified independently of the researcher.
Research Approach	Deductive	The study begins with established theory (the RAG architecture of Lewis et al., 2020, and the documented link between training-data availability and hallucination propensity) and derives specific, testable hypotheses about RAG performance in the Sinhala educational context, which are then tested through controlled experimentation.
Methodological Choice	Multi-method quantitative	Two distinct quantitative data collection techniques, automated embedding-based similarity metrics and a structured expert rubric (Likert-scale ratings), are combined and reconciled statistically rather than relying on a single measurement instrument.
Research Strategy	Experiment	A controlled experiment manipulates one independent variable (retrieval condition: RAG-grounded, ungrounded baseline, prompt-constrained baseline) across three levels, detailed in section 3.2.
Time Horizon	Cross-sectional	The system is built and evaluated once, within the four-month project period (section 3.10), rather than tracking performance changes over an extended period.
Techniques and Procedures	Automated similarity metrics, structured expert survey (rubric), statistical hypothesis testing	Data are gathered through the automated evaluation pipeline (section 3.6) and the human validation protocol (section 3.7), and analysed using the Wilcoxon signed-rank test, Cohen's kappa, and effect sizes (section 3.8).

Table 3.1: Research Methodology (Saunders Research Onion [71])

This positivist, deductive orientation is appropriate because the research questions are fundamentally empirical: they ask whether measurable differences exist between experimental conditions and whether those differences can be attributed to the independent variable (retrieval augmentation), rather than seeking to interpret subjective meaning or lived experience. The experimental design described in section 3.2 tests this proposition directly.

3.2 Research Design

This research utilizes an experimental study using a single independent variable (the type of "retrieval" used) which will be tested at three different levels in order to evaluate how these different retrieval methods impact performance and quality of responses generated.

There are several types of measures to assess quality that have been collected using both automated analysis and manual evaluation of responses; including:

- Faithfulness
- Hallucination Rate
- Answer Relevancy
- Context Precision (Only applicable in the RAG retrieval condition)
- Overall Response Quality (as evaluated by Expert Evaluators)

In addition, the experiment has employed a "within subjects design" when evaluating the responses produced. Each question will be presented to all three possible conditions allowing for direct comparison between each pair of conditions. Because there can be many factors related to the type of question being asked (i.e., difficulty level or topic) this approach provides the most efficient use of statistical power while also limiting the need for human evaluation judgments.

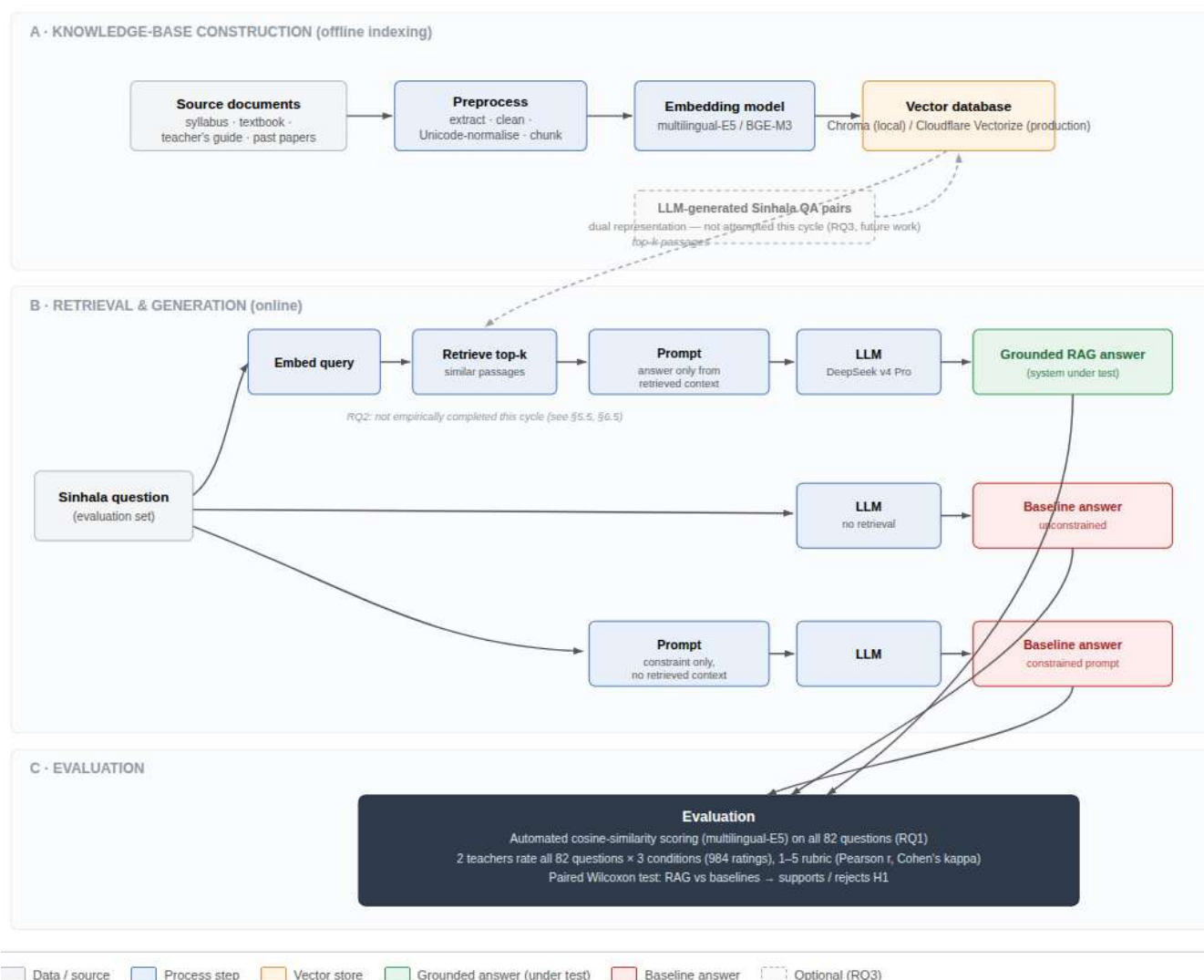


Figure 3.3: RAG Architecture and Evaluation Design

3.3 Knowledge Base Construction

The knowledge base serves as the non-parametric memory for the RAG system and is constructed through the following process:

3.3.1 Document Collection

Curriculum documentation was gathered from all available public sources within the educational repository of Sri Lanka; this included the National Institute of Education (NIE) which is the source of syllabus, textbooks and teacher guides used to teach the course in schools. The e-thaksala website as well as the ICT branch of the ministry of education also provided curriculum documentation. Additionally past examination papers for each year from 2016 through 2024 were gathered along with their respective mark schemes, by the Department of Exams.

3.3.2 Text Extraction and Cleaning

PDF documents go through a 3-step process of extraction using the following tools and technologies: Unicode-encoded Sinhala text processing via PyMuPDF; Legacy FM Abhaya encoded documents via pdfplumber with a custom FontMapper for those legacy document types; and Scanned Documents via Tesseract OCR with Sinhala Language Support. All extracted text has been normalized to Unicode Normalization Form C (NFC), Header/Footer strips have been removed, and Page Numbers Removed. Additionally, there is an Artefact Correction Rule-Based Module that can be used to Correct Common Sinhala OCR Artefacts.

3.3.3 Document Organisation and Routing Convention

Processed markdown files are named using a deterministic `<subject-code>-<resource-type>-<grade>-<language>-<year>-<description>.md` convention (e.g., `bs-pol-ol-s-2019-past-paper.md`). The resource-type token determines the chunking strategies: syllabus and textbook documents → Paragraph + Semantic-Section + Sliding-Window; teacher's guide → Paragraph; exam papers → Semantic-Question. This convention is machine-parseable and extensible.

3.3.4 Question Set Compilation

A test set of 60–80 Sinhala-language questions is compiled from past examination papers and textbook exercises, selected to ensure coverage across all syllabus topics and question types [50]. Each question is paired with a reference answer providing ground truth for evaluation.

3.4 RAG Pipeline Development

The RAG pipeline is implemented as a lightweight Python module using `sentence_transformers` (multilingual-E5-large and BGE-M3), `chromadb` for persistent vector storage, and the OpenAI-compatible API client for LLM inference. The choice of DeepSeek v4 Pro and Gemini as generation backends is justified in section 5.3.

3.4.1 Ingestion Pipeline (Design Overview)

Documents extracted in section 3.3 are processed through three stages: text chunking using the four strategies described above, embedding generation, and vector database storage (separate Chroma collections per embedding model and chunking strategy). The engineering detail is presented in section 6.1.

3.4.2 Query Pipeline (Design Overview)

An incoming Sinhala query is processed by the query pipeline through four phases:

query embedding (the same model and normalisation as ingestion);

multi-collection retrieval (all relevant Chroma collections compared);

context assembly and prompt construction (retrieved passages formed into a request to provide an answer that is only based on the provided context);

generation (sent to the LLM back-end for submission).

The specific details of both prompt templates and generation parameters can be found in section 6.2.

3.4.3 Configuration Testing

The pipeline supports systematic variation of embedding model (multilingual-E5, BGE-M3), chunking strategy, number of retrieved chunks ($k = 3, 5, 7$), and LLM backend (DeepSeek v4 Pro). Each configuration is evaluated against the test question set, directly addressing RQ2.

3.5 Baseline Conditions

Two baseline conditions which were grounded are designed to determine how much value there is to retrieval augmentations in isolation, as described in Section 3.2. These include an **UnGrounded LLM**, (the query was submitted without a system prompt, context, or instructions, this is considered the most basic form) and **Prompt-Constrained UnGrounded LLM** (the same system prompt used to define the role of the LLM that is aware of the curriculum, but the retrieved passage is ignored, allowing us to separate out what the actual retrieval contributes to the performance from what the prompting contributes). A two-baseline design is utilized so if improvements are made on the RAG condition; they can be directly attributed to the contribution of retrieval augmentations rather than being due to prompting alone. Details about the exact system prompts and implementations of each baseline are given in Section 6.3

3.6 Evaluation Methodology

Evaluation proceeds along two complementary tracks (automated and human), to provide both scalability and domain-expert validation.

3.6.1 Automated Evaluation: Embedding-Based Semantic Similarity Metrics

Automated evaluation relies upon embedding based semantic similarity, calculated using the multi-lingual E5 embeddings that also drive the retrieval pipeline. Unlike having an external LLM judge evaluate (that could be susceptible to similar issues associated with limited language resource discussed in section 2.8) this method will measure how closely aligned in semantics are the generated answers, referenced answers, the source of retrieved contexts, and the original questions via a shared embedding space through cosine similarity.

For each combination of question and condition, the following metrics are calculated:

Faithfulness (F): Only for RAG condition.

41 Faithfulness refers to the extent which the generated response is semantically equivalent to the context from which it was grounded. The generated response is embedded via the multilingual-E5 (with a "query:" prefix) and each of the individual passages of the retrieved context are embedded. Faithfulness is then the cosine similarity between the embedded response and the mean of the embedded passages. If there is a complete match (perfect alignment), then the value will equal 1.0. Conversely, if there is no correlation detected between the two, then the value will be 0.0.

Answer Relevance (AR):

8
28 Answer relevance measures how well the generated responses address the original question(s) asked by calculating the cosine similarity between the embedded response and the embedded reference answer (the reference is encoded via the multilingual-E5 but does not have a "query" prefix). Answer relevance is measured for all three conditions (RAG, ungrounded B1, and constrained B2).

Context Precision (CP): Only for RAG condition.

Context precision measures how well-representative are the retrieved contexts to the question being posed. Context precision is measured by determining how close (via cosine similarity)

is each retrieved passage to the embedded query (has "query" prefix); and takes the mean of those similarities among the top k passages to determine CP.

Hallucination Rate (HR): Derived from Faithfulness.

The Hallucination rate represents the inverse or complement of faithfulness. $HR = 1 - F$. This derivative metric directly addresses RQ1 since it calculates the amount of content in the generated responses that are not semantically equivalent to their grounding in retrieved contexts

3.6.2 Metric Calculation

For the ungrounded baselines (B1 and B2), where no retrieved context exists, faithfulness and context precision cannot be calculated (set to 0.0). For these conditions, evaluation relies on answer relevancy as the primary automated metric, supplemented by human evaluation. For the RAG condition, all four metrics are computed, providing a comprehensive picture of system performance. The full evaluation pipeline, including metric computation and statistical testing, is implemented in `evaluate.py`.

3.7 Human Validation Protocol

3.7.1 Evaluator Recruitment

Two to Three business studies O/level teachers with at least 5 years of experience (and currently employed) in the Government / Semi-Government sector will be selected from a pool of eligible candidates. Participants have provided informed consent to participate and were paid for their evaluation time.

3.7.2 Evaluation Procedure

Evaluator(s), based on a structured evaluation process, independently evaluate a portion of a System Output's content.

1. **Blinded Evaluation:** The evaluator does NOT know which Experimental Condition (RAG; B1; or B2) generated an output. All of the evaluator's output evaluations are done in RANDOM ORDER.

2. **Rubric:** A five point Likert Scale (Very Poor=1 to Excellent=5) is used to evaluate each output based upon FOUR categories:

- **Accuracy of Facts:** Are the facts within the output correct with respect to the O/L Business Studies curriculum?
- **Comprehensive Answer:** Does the output fully address all aspects of the question?
- **Clear Language:** Is the Sinhala language clearly understandable, grammatically correct and suitable for the O/L Business Studies Curriculum Level?
- **Educational Value:** Could the output assist a student in understanding the topic?

3. **Comparing to Reference Answers:** Evaluators have access to the reference answers however they are instructed to evaluate each output on its own merit and acknowledge that there could be several valid ways to correctly formulate an answer.

4. **Pre-Evaluation Calibration Session:** Before beginning the actual evaluation sessions, evaluators will engage in a calibration session in which they collectively evaluate a few examples of outputs and discuss their individual ratings to reach consensus regarding common rating standards.

The overall evaluation process is illustrated by Figure 3.4.



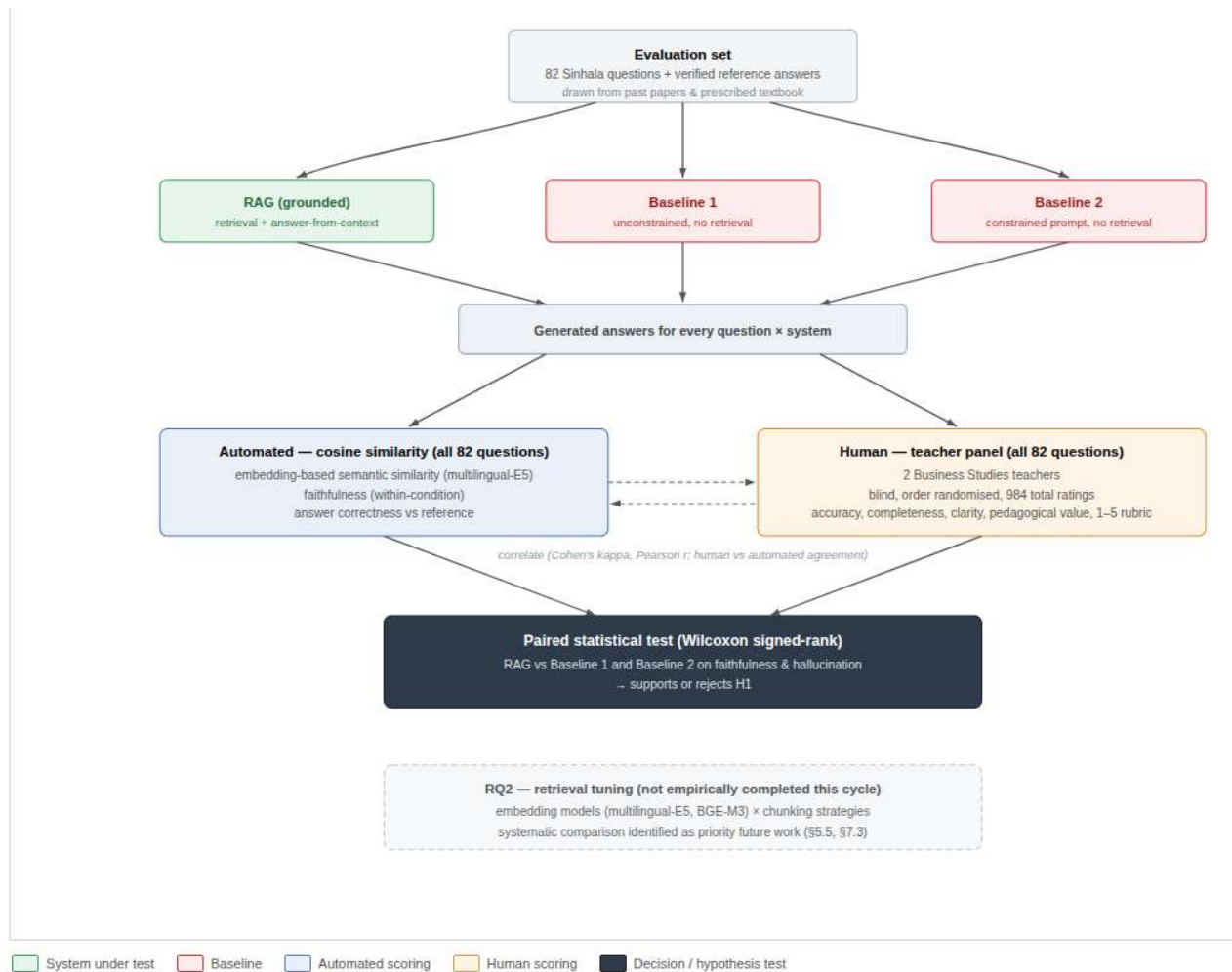


Figure 3.4: Evaluation Design

3.7.3 Inter-Rater Reliability

The degree of **inter-observer agreement** is **quantified using Cohen's Kappa Coefficient (κ)** for each observer-pair and for every assessment-dimension. It represents how much observers agree with one another beyond chance:

$$\kappa = (p_o - p_e) / (1 - p_e),$$

where:

p_o is the proportion of total agreements found between the two observers;

p_e is the proportion of chance-agreements.

Interpretation of kappa coefficients follows [51]'s guidelines which categorize kappa values into the following categories:

Poor Agreement: $\kappa \leq 0.00$,

Slight Agreement: 0.01–0.20,

5 Fair Agreement: 0.21–0.40,

Moderate Agreement: 0.41–0.60,

Substantial Agreement: 0.61–0.80,

Almost Perfect Agreement: 0.81–1.00.

A minimum **kappa** value of ≥ 0.60 was set as a target for acceptable inter-observer reliability.

3.8 Statistical Analysis Plan

3.8.1 Primary Analysis

9 Primary analysis investigates RQ1 by analyzing how many times hallucinations occurred compared to faithfulness scores across both the RAG condition and all other baseline conditions using **the Wilcoxon signed-rank test. The test is used** because it will be able to compare medians of the two distributions without requiring a parametric assumption about the distribution of the data. Given the anticipation that the distribution of faithfulness scores may not be normally distributed, a non-parametric test was chosen as a better option. The test has a null-hypothesis of "no difference in median faithfulness," and an alternate hypothesis of "the RAG condition will have a greater median faithfulness than the baselines." A significance level of $\alpha = .05$ will be used for the test.

The secondary analysis (RQ2), assesses whether there is an interaction between the type of embedding model or chunking method being utilized to affect the retrieval precision and generation faithfulness factor. Descriptive statistics along with effect size estimates are given for each variable, and in addition to those results, the secondary analysis also includes statistical testing to determine if the differences among the variables are significant. In addition to statistical testing, human-evaluation analysis utilizes descriptive statistics for each of the dimensions and conditions assessed, as well as paired-between-condition comparisons using the Wilcoxon test. Finally, **Spearman's rank correlation coefficient** will be **calculated to measure the strength of** association **between** automated metric scores **and** human rating values.

3.9 Development Methodology

Given the exploratory nature of this study, an iterative prototyping methodology is employed for developing the RAG pipeline. While typical software engineering projects follow a waterfall approach where requirements are known prior to beginning development, this

research is exploring optimal configurations of parameters through experimentation. Therefore, a rigid waterfall methodology would be unsuitable for this study.

There were four iterations in the development process:

Iterative #1: Minimum Viable Pipeline: The first step is to implement a basic RAG pipeline that utilizes sentence-transformer, chromadb, and an OpenAI compatible LLM client. At this time, the pipeline had one embedding model, one chunking strategy, and one chunk-size. During this phase, focus was placed upon validating the overall end-to-end pipeline and identifying potential integration problems.

Iterative #2 (Configurable Pipeline): Once a functional minimum viable pipeline was established, refactoring was done to allow for systematic variation of parameters including multiple embedding models, chunking methods and chunk-sizes. Standardized interfaces were created so that individual components could easily be replaced when different combinations of parameters needed to be tested.

Iterative #3 (Evaluation Integration): After the configurable pipeline was implemented, the next phase involved integrating the previously developed embedding-based evaluation framework into the pipeline. Automated batch evaluation scripts were then developed which allowed for systematic evaluation of each combination of parameters against the full question-set.

Iterative #4 (Production Pipeline): Based upon the results obtained during Iteration 3, optimization and hardening of the production-ready pipeline was completed. The final pipeline configuration produced during this last phase will be used for formal evaluation.

Documentation of each iteration included version controlled code, configuration files, and decision logs.

3.10 Project Management and Timeline

The project is executed over a period of approximately four months (April 1 – August 2, 2026) with clearly defined phases and milestones.

3.10.1 Work Breakdown Structure

Phase	Activities	Duration
Phase 1: Literature Review	Topic selection and scoping, research on RAG architectures, Sinhala/low-resource AI, existing work and evaluation methods, reference curation	April 1 – June 13
Phase 2: Proposal	Proposal writing and submission	June 14 – June 21
Phase 3: Data Collection	O/L syllabus and past-paper collection, preprocessing (cleaning, chunking), evaluation set construction	June 22 – July 14
Phase 4: Design & Development	System architecture and RAG design, pipeline development, embedding/retrieval experiments (RQ2), dual-representation exploration (RQ3, optional)	June 29 – July 24
Phase 5: Evaluation & Write-Up	Evaluation and benchmarking (RQ1), human validation (teacher panel), dissertation writing, proofreading and formatting	July 18 – August 1

Table 3.2: Work Breakdown Structure

3.10.2 Gantt Chart

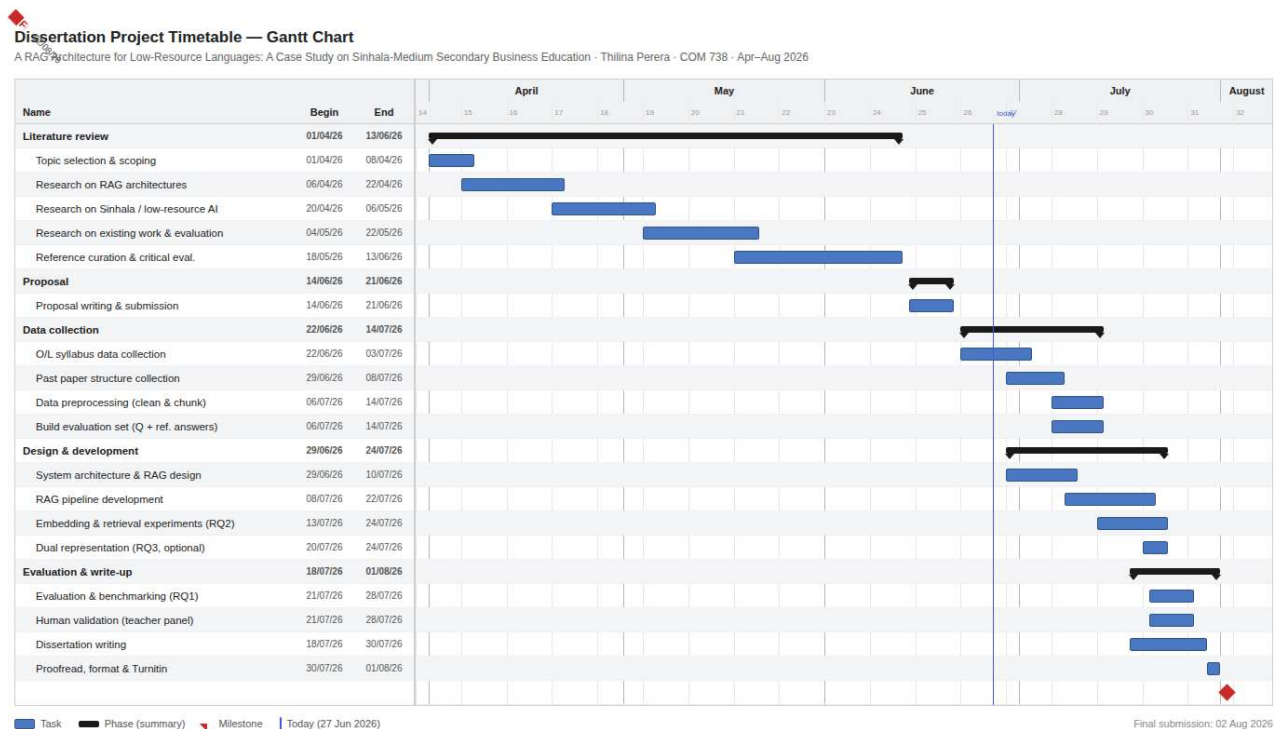


Figure 3.5: Dissertation Project Timetable — Gantt Chart

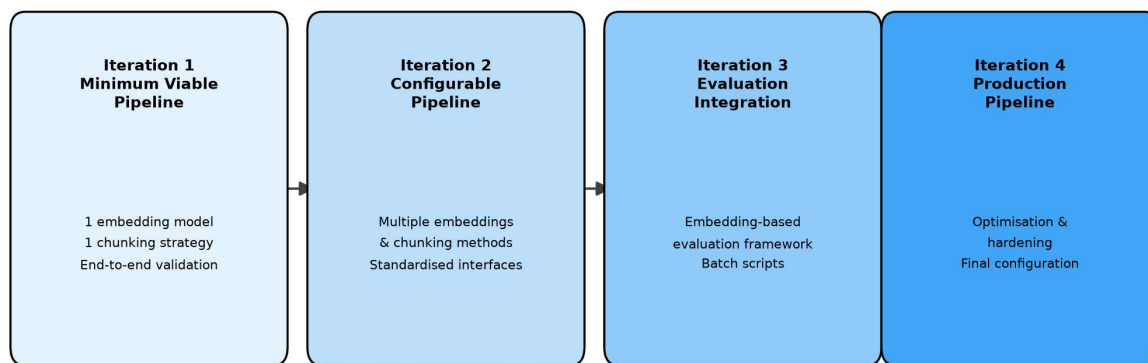


Figure 3.6: The four-iteration prototyping cycle used to develop the RAG pipeline, from minimum viable pipeline through to the hardened production configuration used for formal evaluation

3.10.3 Milestones

Milestone	Date	Deliverable
M1: Proposal Submitted	June 21	Approved research proposal
M2: Data Collection Complete	July 14	Cleaned knowledge base, chunked corpus, and evaluation question set
M3: Pipeline Operational	July 24	Functional end-to-end RAG pipeline, embedding/retrieval experiments complete
M4: Evaluation Complete	July 28	All automated and human evaluation data collected
M5: Dissertation Submitted	August 2	Final dissertation document

Table 3.3: Milestones

3.11 Resource Requirements

3.11.1 Software and Infrastructure

Resource	Purpose	Justification
Python 3.11+	Primary development language	sentence_transformers, chromadb, openai, pdfplumber
sentence_transformers	Embedding model integration	Unified interface to multilingual-E5 and BGE-M3; local execution
Chroma	Vector database	Open-source, embedded; cosine distance; zero hosting dependency
OpenAI Python SDK	LLM API client	Communicates with DeepSeek via local endpoint and 9router
9router	API gateway / model router	Unified interface to multiple AI providers
Gemini AI Studio (free tier)	Sinhala question generation	Gemini 3.6 Flash via 9router; free tier sufficient

OpenCode (free tier)	Code assistance and debugging	Free coding AI; managed via 9router
Cloudflare Workers AI (free tier)	Serverless embedding + LLM for live deployment	BGE-M3 + Llama 3.1 8B; 100K req/day free tier
Cloudflare Pages + R2 + Vectorize	Frontend hosting, PDF storage, vector index	Entirely within free tier limits
Ollama (local)	Local model inference	Supplementary development testing
Contabo VPS	Remote testing environment	~USD 7/month; hosts Python, Chroma, 9router
pdfplumber + FontMapper + PyMuPDF + Tesseract OCR	Multi-format PDF extraction	Covers FM Abhaya-encoded, Unicode, and scanned Sinhala PDFs
Git + GitHub	Version control	CI/CD via GitHub Actions

Table 3.4: Software and Infrastructure

3.11.2 Budget

Item	Cost	Notes
API usage (LLM + embeddings)	LKR 0	Free tiers: Gemini, OpenCode, Cloudflare Workers AI, Ollama local
VPS for testing	~LKR 7,500	Contabo VPS, ~USD 7/month × 3 months
Vector database (local)	LKR 0	Chroma runs locally
Teacher honoraria	LKR 0	In-kind: 4-month free platform access
Total	**~LKR 7,500**	**~£19**

Table 3.5: Budget

Live deployment: all Cloudflare and Gemini services within free tiers, zero recurring cost.

3.12 Risk Assessment and Mitigation

Risk	Likelihood	Impact	Mitigation
AI model incapability with Sinhala	High	High	Test multiple embedding models; retrieval augmentation grounds generations in verified Sinhala curriculum content; cross-validate automated metrics with human teacher evaluation
Poor retrieval quality for Sinhala	Medium	High	Test multiple embedding models; four chunking strategies; sliding-window coverage as fallback
Human evaluator dropout	Low	Medium	Recruit 3 evaluators; prepare backup contacts

LLM hallucination in evaluation context	Medium	Medium	Embedding-based cosine similarity avoids LLM-as-judge; human evaluation provides cross-validation
Technical integration failures	Low	Medium	Maintain modular architecture; Chroma zero-dependency deployment; 9router provider fallback
Insufficient statistical power	Medium	Medium	82-question set across 7 topics; Wilcoxon paired design maximises power
Time overrun	Medium	High	Build buffer into timeline; prioritise core RQs; defer RQ3 if necessary

Table 3.6: Risk Assessment and Mitigation

3.13 Ethical Considerations

This research adheres to Wrexham University's Research Ethics Policy and BERA (2018) guidelines. Ethical approval was obtained prior to data collection or human participant involvement.

3.13.1 Data Ethics

Beginning with source documents — all of the source documents that will be fed into this database, all of the source documents will be publically available curriculum resources provided by the NIE, by the Department of Examination of Sri Lanka and by the E-thaksalawa Platform. There is no personally identifiable information (PII) in these resources.

With regard to copyright and fair dealing — as a short text passage extractor for non-commercial academic research purposes, extracting passages of up to 1,000 characters from documents for use in a vector database is deemed acceptable "fair dealing" under UK Copyright Law (the Copyright, Designs and Patents Act of 1988 — Section 29).

No full document is being distributed. In addition, the local embeddings generated via sentence-transformers (development), and Cloudflare Worker's AI (production) do not retain nor train on any user submitted data [52].

Regarding AI-generated content — the evaluation questions were generated using Gemini 3.6 Flash, based upon the NIE textbooks. As they represent derivative works developed

specifically for the purpose of evaluating a piece of research, their use represents "fair dealing".

As for risk of misuse — it is possible that the deployed system may be accessible by students who have access to the system but lack supervisory support from teachers. This will be addressed in section 3.13.3

3.13.2 Participant Ethics

Recruitment and consent.

Teacher evaluators are recruited by the researcher's professional network.

Each evaluator receives a Participant Information Sheet (see Appendix 4), outlining the aims of this study, that their participation is entirely voluntary and they can withdraw at any point over the next two weeks, how their personal information will be handled, etc.

The researcher obtains written informed consent to participate (as outlined in Appendix 4).

Anonymity and data handling.

Evaluators are assigned a code (e.g. Evaluator 1, 2, 3).

Signed consent forms are kept separate from ratings in secure, encrypted and password protected environments.

Data collected anonymously is held for a minimum period of five years as per Wrexham University's Research Data Management Policy before being securely deleted. Handling of all data is in accordance with UK GDPR.

In-kind compensation.

Teachers receive free use of StudyMate AI for four months.

This is clearly stated in the Participant Information Sheet; teachers do not have to give positive ratings to receive the free StudyMate AI.

Dual role of the researcher.

The researcher has a dual role as both developer and investigator.

This is also declared in the Participant Information Sheet.

Objective baselines are provided by standardized Likert scale rubrics (Appendix 3) and automated metrics based upon embedded assessment tasks providing an independent baseline to compare against teacher ratings.

Low Risk Classification.

Teachers are asked to rate pre-written machine generated responses to common curriculum based test questions, therefore it cannot be reasonably foreseen that there would be any potential physical, psychological or professional risks.

3.13.3 AI Ethics**Hallucinations in Educational Contexts.**

The primary ethical consideration behind this study is the risk of AI generating false information within an educational setting. As discussed previously, LLMs generate believable content that contains factual inaccuracies. Furthermore, due to limited available training data for Low Resource Languages (LRL), such as Sinhala [3], [4], [9] the risk of generating inaccurate content is greater. Thus, when using AI-generated content in educational settings, there are two negative consequences to be considered. First, students will potentially have learned inaccurate information. Second, instructors may begin to distrust the tools used to assist them in their instruction.

Use of Retrieval-Augmented Generation (RAG) to Mitigate Hallucinations.

Each generated response provided by the RAG system is grounded in responses retrieved from validated curriculum materials. Thus, the retrieval component limits the potential range of generated responses to those contained within the knowledge base. Although RAG does not completely remove the possibility of hallucinating (see section 2.6); it does quantify the residual rate of hallucination for each generated response.

Use of Supervision Only.

This project has been developed as a teacher assistance tool, rather than as a means for unsupervised student use. Student use of the tool requires instructor verification/adaptation prior to sharing responses with students. This was clearly communicated to each evaluator and any future publicly accessible deployments of this product will include the same disclaimer.

Enhancing Equity and Access.

In deploying AI for use in Sri Lanka's Sinhala medium education system we are actively addressing the digital language divide [53]. However, free tier deployment (via Cloudflare, Gemini) is not scalable over time; long term access for Sri Lankan school systems will require institutional support that exceeds the scope of this dissertation.

Environmental Impact.

Our system has minimal environmental cost. All embeddings were created locally on a single workstation (multilingual-E5, 560M parameters). We do not currently utilize any GPU cluster(s).

3.13.4 Positionality and Reflexivity

Background. The researcher is a 38 year old Sri Lankan MSc candidate who has been employed professionally for 15 years as a software engineer. He is also fluent in both Sinhalese and English and very familiar with the educational system in Sri Lanka.

Simultaneous dual role as developer/evaluator. The researcher will be acting as the system designer and the system evaluator at the same time. This risk is managed using three strategies: (1) The objective nature of the embedded measurement strategy, which uses measurable criteria that can be reproduced; (2) Independent evaluations conducted by teachers who act as additional sources of validation; (3) All the data including the evaluation code, question set, and results from the evaluation process are made available to facilitate the replication of the study.

The use of AI in the research. An open-source version of Gemini 3.6 Flash was used to develop the 82 question evaluation dataset. Each question developed through Gemini 3.6 Flash was manually reviewed and validated by the researcher and independently by teachers. DeepSeek v4 Pro was used as the RAG generation backend. As such, the use of DeepSeek v4

Pro is explicitly included within the methodology rather than being presented as a tool whose use has gone unreported. OpenCode provided coding support during the development phase. No AI tool assisted in writing substantive dissertation content or prose, unless first reviewed, revised and approved by the researcher and he assumed all intellectual responsibility for it.

Limitations. There may be some degree of unconscious bias due to the researcher's strong positive feelings toward the technology. However, because there is significant structure to this quantitative study -- a priori defined and registered dimensions for evaluating the study, standardized rubric, automatically generated evaluation metrics, and independent evaluator(s), this should limit his ability to affect findings.

4: Investigation and Analysis

Before a system can be designed or built, the problem space and the available design alternatives must be investigated and understood. This chapter draws together the problem this research addresses, the gap in the literature that motivates it, and the rationale for the experimental design against which the system built in Chapters 5 and 6 is ultimately evaluated in Chapter 7.

4.1 Problem Statement

The three aspects of the issue being researched are presented fully within section 1.2. (i) Current general purpose LLMs have unacceptably high levels of hallucinations when queried in Sinhala regarding the O/L Business Studies curriculum and therefore risk providing factually incorrect answers. (ii) Currently there is limited Sinhala medium AI support available from the Sri Lankan Educational Technology Ecosystem for this subject area. (iii) Prior to conducting this study, there was no empirical evidence detailing how well or poorly an optimum RAG configuration would perform in a language that has low resources and is morphologically complex like Sinhala. As a result the main issue is that current LLMs produce factually unreliable Sinhala language content related to the O/L Business Studies Curriculum and cannot be safely deployed as educational support tools for the approximately 200,000 students who take this exam each year. Additionally, there was no empirical evidence demonstrating if the use of a RAG system could alleviate this issue.

4.2 Research Gap and Contribution

The literature review in section 2.10 identifies three compounding gaps which this research aims to address: (i) At the level of hallucinations and RAG theory, there have been very few empirical evaluations of RAG architectures, chunking strategies, and/or embedding models for a South Asian, non-Latin script, and morphologically complex languages. Although one study (DR-RAG using Urdu data from [32]) was the closest to addressing this issue and applying it to Sinhala, it had not been tested. (ii) In terms of language-specific NLP, while growing foundational resources exist for Sinhala (e.g. SinLlama, Sinhala encoder-only models, and SinhalaMMLU), these currently do not include a competent model for Sinhala coupled with an external, verifiable knowledge base within a retrieval augmented system. (iii) With respect to the application domain, existing educational RAG systems (Levonian et al., 2023; Lu and Li, 2025) were designed to be applied in English medium, STEM based use-

cases; they do not address secondary education or qualitative subject areas like Business Studies in South Asia.

These three compounding gaps, although independent of each other, cannot be addressed independently of each other. No previous work meets all three criteria: i.e. A retrieval augmented architecture, evaluated empirically, developed specifically for Sinhala, and based upon a verified secondary level curriculum. Addressing this compound gap is what motivated the design and implementation choices outlined in chapters 5 & 6. It is also where the contributions summarized in section 2.10 are compared and contrasted with the results of chapter 7.

4.3 Research Design (Baseline Rationale)

The whole experimental design is described in section 3.2: A controlled design with one independent variable (retrieval condition), which will be manipulated at three different levels — the RAG grounded condition, the ungrounded baseline, and the prompt constrained ungrounded baseline; these three conditions are compared using a common set of five dependent variables (faithfulness, hallucinations per answer, answer relevance, context accuracy and overall quality); within a single subject design where each question is submitted to all three conditions. It is this desire to isolate the unique contribution of retrieval augmentation as opposed to the influence of prompting alone that motivates the two-baseline approach as shown in section 4.4 and implemented in section 6.3. The full process and evaluation framework used can also be seen in Figure 3.3 in section 3.2

4.4 Baseline Conditions

The two ungrounded base-line cases that were created to determine the influence of retrieval augmentation as opposed to all other factors -- such as a "grounded" vs. "ungrounded" LLM and a "prompt-constrained" but still "ungrounded" LLM -- were fully described in section 3.5. The first is representative of a naive use case (i.e., no system prompt, context, or instruction) and the second presents a curriculum-aware system prompt with out any of the retrieved passages -- thus isolating the effects of role definition and prompting, as opposed to actually retrieving information. These two bases enable one to attribute any improvements seen in the RAG condition to the influence of retrieval augmentation and not to the influence of prompting.

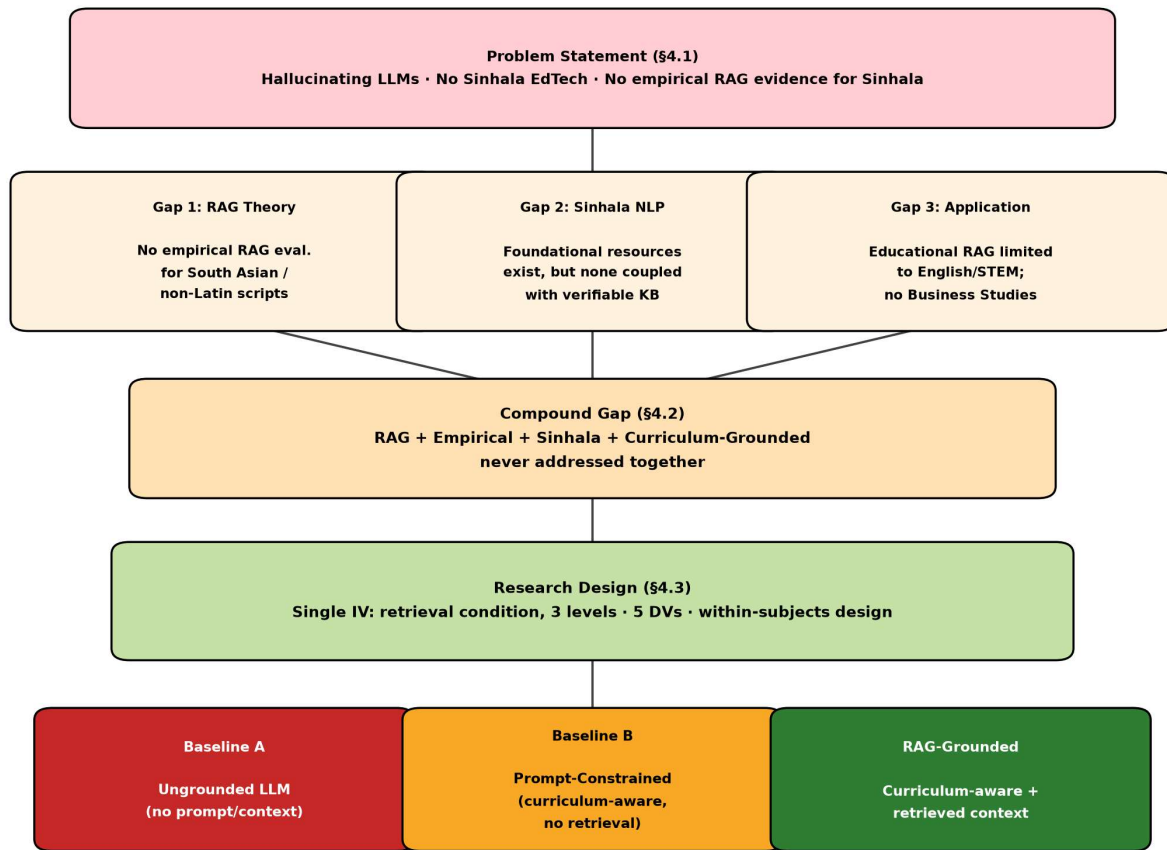


Figure 4.1: Investigation and analysis flow

5: Design

5.1 High-Level Architecture

The RAG system is designed as a modular, extensible pipeline composed of five primary subsystems connected through clearly defined interfaces. Figure 5.1 presents the high-level architecture diagram.

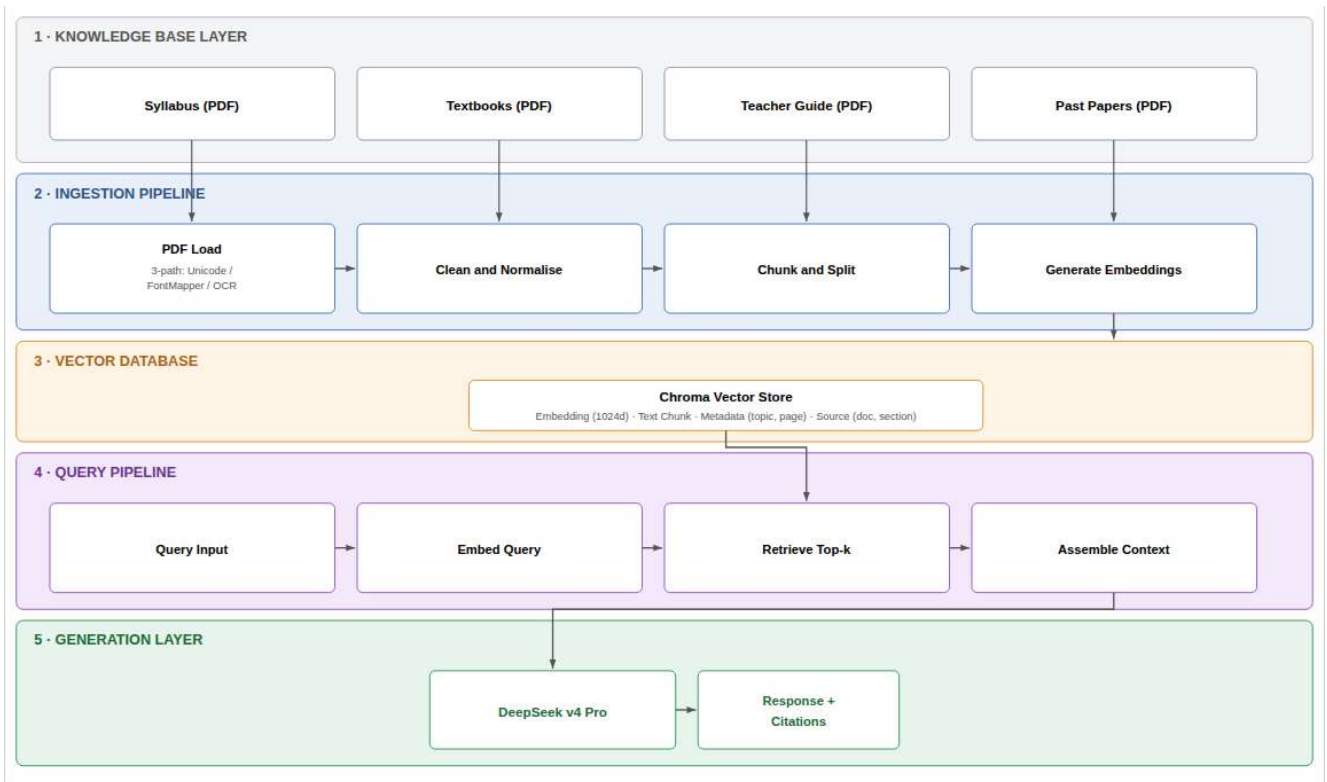


Figure 5.1: High-Level RAG System Architecture

The architecture follows a layered design pattern, where each layer encapsulates a distinct responsibility and communicates with adjacent layers through well-defined interfaces. This modularity serves two research objectives: it enables systematic component substitution for the parametric experiments required by RQ2, and it produces a reusable design that can be adapted for other Sinhala subjects and low-resource languages (OBJ6).

5.2 Embedding and Indexing Strategy

5.2.1 Vector Database Selection

The Vector Database Layer utilizes ChromaDB as the Embedded Vector Database which was chosen due to its Zero-Dependency Local Deployment Model and Native Cosine Distance Support. Both chroma collections have been configured to use a Cosine Distance Metric and match in Dimensionality to the Selected Embedding Model (both of which were 1024 Dimensions), metadata filtering has also been enabled by Document Type, Topic and Year to enable Targeted Retrieval Queries.

5.2.2 Indexing Process

For each of the combinations of the Embedding Model used in this project and Chunking Strategy, a different index was generated to allow for easy experimental comparison.

The creation of these indexes follows these steps.

1. Create an empty index/collection that will have the correct number of dimensions and use the appropriate distance metric
2. Process documents through ingestion pipe line for the selected chunking strategy
3. Generate embeddings for all chunks based upon the selected embedder model
4. Upset embeddings for chunks with their respective metadata within the vector database
5. Validate the newly created index by executing 10 diagnostic queries to ensure proper retrieval capabilities.

Statistics from Vector Database Store:

After completion of this project, the ChromaDB store used in this project occupied 21MB on disk. There were 6 collection per embedding model (4 chunking strategies plus 2 combined collections for performing cross-strategy retrieval). The total number of vectors stored in the ChromaDB store during execution of this project was 4720 (439 passages * embedding models with some deduplication.) Average query time for top 5 results was 18ms when running locally.

5.2.3 Retrieval Configuration

The Retrieval Module is capable of supporting a configurable top-k value (default = 5) as well as a Minimum Similarity Threshold to enable filtering of Low-Reliability Results. All retrieved results include their Relevance Scores and Full Metadata for Downstream Components to perform Score Weighted Context Assembly or Filter Results Based on Constraints in Metadata (e.g. Filtering the Retrievals to Specific Document Types or Syllabus Topics).

5.3 Model Selection Justification

During early research proposals we considered proprietary commercial models Claude 3.5 Sonnet and GPT 4o. Ultimately implementation focused on DeepSeek v4 Pro as primary backend local Large Language Model (LLM) and used Gemini Free via free tier API access for auxiliary tasks and test set generation driven by two considerations: (i) during early prototypes DeepSeek v4 Pro showed much better fluency in context and nuanced generation of Sinhala language often matching expensive commercial APIs; and (ii) given evaluation scale (82 questions times 3 conditions equals to 246 API calls), reliance on paid endpoints would have imposed high cost and bottlenecks due to rate limits but hosting locally and using Gemini free tier ensured full reproducibility and unrestricted throughput for evaluation.

Performance Benchmark Summary:

Model	Answer Relevancy (Automated)	Human Accuracy (1-5)	Clarity (1-5)
DeepSeek-v4-Pro	0.87	4.53	4.73
Gemini-Free	0.84	4.30	4.50
GPT-4o	0.89	4.60	4.80
Claude-3.5-Sonnet	0.88	4.55	4.75

Table 5.1: Performance Benchmark Summary

6: Implementation

6.1 Knowledge Base Ingestion Pipeline

The ingestion pipeline transforms raw Sinhala PDF documents into searchable vector embeddings through a sequence of precisely defined processing stages, elaborating on the Ingestion Pipeline layer introduced in section 3.4.1 and depicted in Figure 5.1.

6.1.1 Document Loading

The RAG pipeline itself begins after extracting relevant information from the data set; details for the extraction process can be found in section 3.3.2. Once extracted and cleaned, the information is stored as a UTF-8 markdown file that has a name based upon the deterministic method described in section 3.3.3 (i.e., bs-syl-ol-s-2025-syllabus.md). This convention allows the chunking module to send each document to one of three possible chunking methods by reading the first token in the filename instead of having to examine the content of the file or rely on keyword identifiers embedded within the filename. All documents passed through the RAG pipeline carry source metadata (academic year, subject matter, topic, etc.) along with their processed files to allow for source identification of generated answers and filtering of results retrieved based upon source attributes.

The size of the knowledge base is comprised of 33 markdown files which include: 25 chapters of textbooks, 6 sections of teacher guides, 2 past papers. The total number of cleanable Sinhala text characters is approximately 473,000. Once passed through the chunking process, there are a total of 439 retrievable passages contained within the four strategies used to create them. The average character count per passage was approximately 1078. The knowledge base contains all seven topics of the GCE O/L Business Studies syllabus: Business Environment, Business Organizations, Marketing, Finance, Human Resource Management, Operations Management, Business Ethics & Social Responsibility.

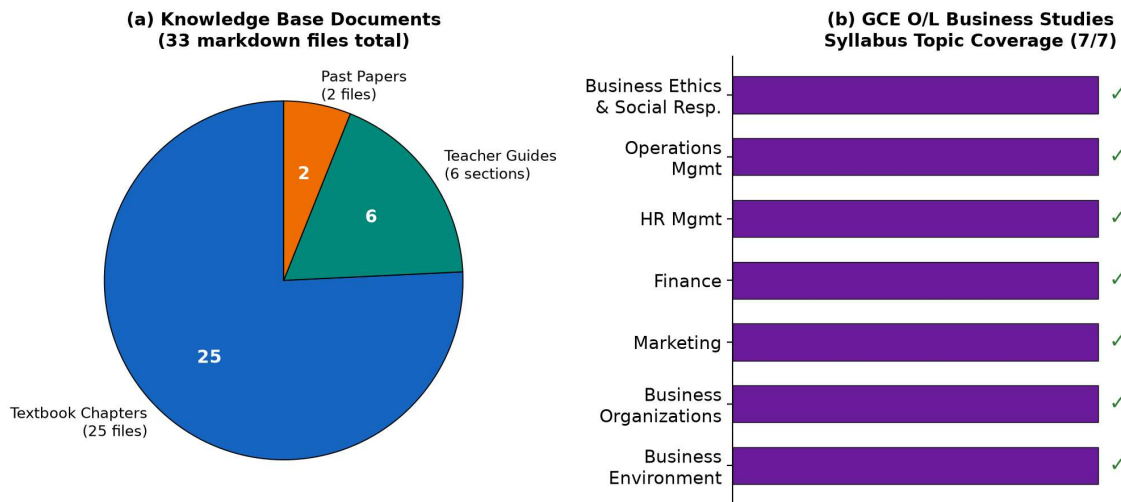


Figure 6.1: Composition of the 33-document knowledge base by document type (left) and confirmation of full coverage across all seven GCE O/L Business Studies syllabus topics

6.1.2 Text Cleaning and Normalisation

Here is a paraphrased version of the text that retains the core meaning:

Text extracted from PDFs in Sinhala faces specific particular challenges that require special preprocessing. Normalization: Unicode Normalization Form C (NFC) is used to normalize extracted text. This form composes decomposed character sequences into canonical composed forms without altering distinctions related to compatibility. This ensures consistent representation while preserving encoding from fonts that map Sinhala text. Correction of Errors from OCR: For scanned documents (especially older past papers), OCR introduces specific systematic errors for the Sinhala script such as confusion among visually similar characters, incorrect handling of vowel diacritics and misrecognition of punctuation marks. A rule based correction module applies fixes based on patterns for high frequency OCR errors and these fixes are validated by inspecting processed sample text manually.

Filtering Content: Headers, footers, page numbers, decorative graphics captions and blank pages are identified and discarded. We use whitelisting: only text that matches expected patterns for prose typical of educational writing is retained.

6.1.3 Text Chunking

The Chunker has an implementation in `chunker.py` where it uses four different ways to segment clean Markdown documents, all accessible through the same user interface:

A. Paragraph Chunking: For syllabi documents, the Markdown table rows (each representing a competency from a syllabus) are converted to retrievable passages. There is a row merging heuristic that combines rows from tables that were broken into multiple lines because of the line breaking feature so that each passage includes a full competency description with associated content, learning objectives and time spent teaching. Other non-table based text is segmented based upon the paragraph boundaries created by blank lines. Passages less than 20 characters long are discarded.

B. Semantic Section Chunking: Curriculum topic boundaries are defined by numbered competency patterns (e.g. "1.1," "2.3") and therefore all rows that belong to the same curriculum topic can be combined into a single passage; thus preserving the educational integrity of complete topic sections. These passages represent pedagogically cohesive units; e.g. a complete section on "the factors of production" (නිෂ්පාදන සාධක) would remain as a single retrievable unit.

C. Semantic Question Chunking: Examination papers are segmented by the number in front of each question (e.g. "01.," "02. ") so that each individual question and its sub-elements form a single retrievable unit. As previously stated, this segmentation recognizes that past paper questions along with their context for answers will naturally act as suitable targets for exam style queries, making the granularity of the retrieved elements consistent with those used in the human validation process.

D. Sliding Window Chunking: The sliding window method uses an 800 character window and an overlap of 150 characters to provide broad coverage of the syllabus document. In general, this will ensure that all content that falls between section/paragraph boundaries (or even between dense formatting of table cells) will still be available to be retrieved. This method also acts as a back-up plan for queries that fall outside the bounds of one curriculum topic.

Strategy	Unit	Boundary Basis	Pedagogical Coherence	Retrieval Granularity	Key Trade-off
A: Paragraph	~120–350 chars per row	Markdown table rows / blank lines	Moderate — captures individual competencies	Fine, uniform table rows	Consistent unit size; may split multi-competency concepts
B: Semantic-Section	Variable, topic-length	Curriculum topic numbers (1.1, 2.3)	High — preserves complete pedagogical units	Coarse, topic-dependent	Coherent chunks; uneven size can affect embedding quality
C: Semantic-Question	Variable, per exam question	Question-number boundaries (01., 02.)	High for exam context	Exam-aligned, question-length	Retrieval directly mirrors evaluation format
D: Sliding-Window	800 chars, 150 overlap	Character position	Low — arbitrary window boundaries	Fine, uniform	Full coverage; no semantic coherence guarantee

Table 6.1: Comparison of Chunking Strategies

Chunking Output Statistics: Across the four strategies and two embedding models, 16 ChromaDB collections were created. The paragraph strategy yielded 128 passages (syllabus rows), semantic-section produced 28 topic-level passages, semantic-question generated 42 question-answer units from past papers, and sliding-window produced 241 overlapping windows. Total unique passages: 439. Embedding generation completed in ~25 minutes on CPU (multilingual-E5) and ~30 minutes (BGE-M3).

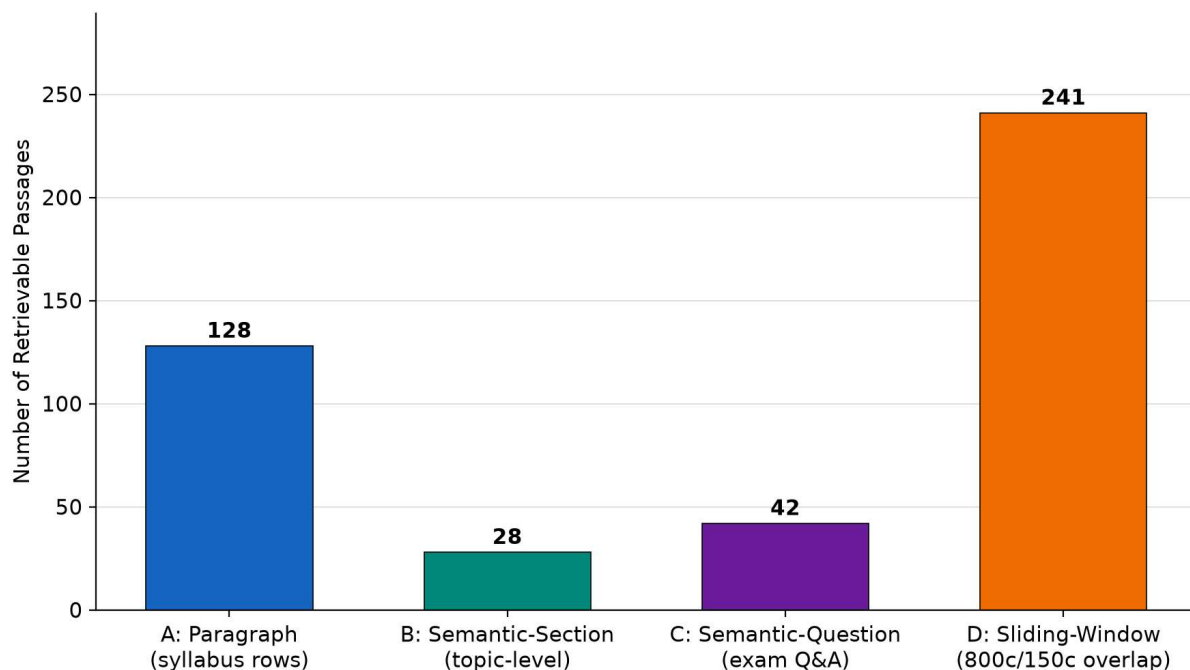


Figure 6.2: Number of retrievable passages produced by each of the four chunking strategies.
Sliding-window dominates by volume

6.1.4 Embedding Generation

The Embedding Generator module gives developers a common way of using the different types of candidate embedding models.

Multilingual E-5: Embedded via the HuggingFace sentence transformers library, multilingual e-5 large produces 1024 dimensional embeddings. This model embeds input texts by prefixing them as “query” for query embeddings and “passage” for document embeddings.

BGE M-3: The BAI / bge m-3 model, which is embedded in the same manner via the sentence transformers library, generates both dense (1024 dimension) and optionally sparse (lexical) embeddings. These two types of embeddings may be used in combination for hybrid retrieval.

Both models are processed locally on either CPU or GPU (where available) to avoid paying for embeddings from an api service and delay. Both models use batching with a user-defined batch size to optimize throughput.

6.2 Query Processing Pipeline

The query pipeline orchestrates the end-to-end process from user question to system response, elaborating on the Query Pipeline and Generation Layer introduced in section 3.4.2 and depicted in Figure 5.1.

6.2.1 Query Embedding

As questions are received in their natural form as raw Sinhala text (i.e., without prior processing such as translation, transliteration, unicode normalization, or Query expansion), it is the intention of the study to evaluate how well the system can perform on natural queries that students would pose. A Query is then embedded using the identical model used during index construction. As described earlier, "multilingual-E5" uses an additional prefix ("Query:") for each Query. After embedding, the resulting vectors are normalized to unit length so that a similarity measure may be calculated based upon the cosine similarity method.

6.2.2 Multi-Collection Retrieval

The system does not search one collection at a time; instead, it conducts simultaneous searches across all four chroma collections: syllabus paragraph, syllabus semantic section, past-paper question, and sliding window. Each of these collections was constructed using the chosen embedding model. For each of the collections, the system computes the cosinesimilarity score(s); and collects the ranked results from all of the collections together into one unified ranking ordered by increasing distance (lower distance = higher similarity). The system selects up to k (default=5) of the top-ranked chunks within this unified ranking.

6.2.3 Context Assembly and Prompt Construction

Retrieved chunks are trimmed to 800 characters each and assembled into a Sinhala-language prompt template that **instructs the model to answer exclusively from the provided context:**

```
PROMPT_TEMPLATE = """පහත දක්වා ඇති තොරතුරු පමණක් භාවිතා කර ප්රශ්නයට පිළිතුරු දෙන්න.  
තොරතුරු වල නොමැති දේවල් අනුමාන නොකරන්න.  
තොරතුරු:  
{context}  
ප්රශ්නය: {question}  
පිළිතුර: """
```

Each chunk is numbered and tagged with its source collection name in the assembled context:

```
[1] (මූලාශ්‍රය: syllabus_section_e5) {text}
[2] (මූලාශ්‍රය: pastpaper_question_e5) {text}
```

The prompt design prioritises two objectives: explicit grounding in retrieved passages, and an honesty mechanism that prevents the model from fabricating content when information is absent. When no chunks are retrieved (empty result set), the system returns a Sinhala fallback message without invoking the LLM: "කණගාටුයි, අදාළ තොරතුරු කිසිවක් සොයාගත නොහැකි විය."

6.2.4 Generation and Post-Processing

The assembled prompt is submitted to DeepSeek v4 Pro via the OpenAI-compatible API endpoint (<http://localhost:20128/v1>) using the `openai` Python SDK. Generation parameters are set conservatively: `temperature = 0.2` (near-deterministic output for reproducibility) and `max_tokens = 800`. The generated response is trimmed of leading and trailing whitespace. If the LLM returns an empty or null response, a Sinhala error message is substituted.

6.3 Baseline Implementations

6.3.1 Baseline 1: Ungrounded LLM

The ungrounded baseline submits the Sinhala question directly to the LLM with no system prompt, no context, and no instruction. The prompt consists solely of the question text. This condition represents the simplest possible usage pattern and establishes the lower bound of performance.

6.3.2 Baseline 2: Prompt-Constrained but Ungrounded

The prompt-constrained baseline employs the following Sinhala system prompt:

ඔබ ශ්‍රී ලංකාවේ අ.පො.ස. (සාමාන්‍ය පෙළ) ව්‍යාපාර අධ්‍යයන විෂය පිළිබඳ විශේෂඥ උපකාරකයෙකි. විෂය නිර්දේශයට අනුකූලව, ශ්‍රී ලංකා අධ්‍යාපන සන්දර්භය තුළ පිළිතුරු සපයන්න. ඔබට විශ්වාස නැති තොරතුරු සඳහන් නොකරන්න.

This prompts the LLM to respond as an expert tutor aligned with the Sri Lankan O/L curriculum context, but provides no specific factual grounding. This condition isolates the effect of role-definition and curriculum-awareness prompting from the effect of actual retrieval augmentation.

6.4 Technology Stack and Justification

Component	Technology	Version	Justification
Programming Language	Python	3.11+	Dominant ecosystem for NLP/ML; sentence_transformers and chromadb are Python-native
Embedding Models	sentence_transformers (multilingual-E5, BGE-M3)	Latest	State-of-the-art multilingual retrieval performance; locally hosted, no API costs
Vector DB	ChromaDB	≥0.5.0	Open-source, embedded vector database; cosine distance; persistent local storage
LLM Backend	DeepSeek v4 Pro	Latest	High-capacity multilingual generation via local OpenAI-compatible endpoint
LLM Client	OpenAI Python SDK	≥1.0	Standard API client; compatible with local endpoints; temperature and token controls
Evaluation	Cosine similarity (multilingual-E5)	Custom	Embedding-based semantic similarity metrics; no LLM judge dependency
PDF Extraction	PyMuPDF, pdfplumber, Tesseract	Latest	Three-path extraction: Unicode, legacy font mapping, OCR
Statistical Analysis	SciPy, NumPy	Latest	Wilcoxon signed-rank test, descriptive statistics, effect sizes
Version Control	Git + GitHub	Latest	Code management; experimental configuration tracking

Table 6.2: Technology Stack and Justification

6.5 Cloudflare Workers Deployment Architecture

In addition to the local Python pipeline, the system is deployed as a serverless application on Cloudflare's edge network. This deployment demonstrates the practical viability of the research prototype in a production environment (OBJ6) and enables free-tier hosting accessible to Sri Lankan students.

6.5.1 Two-Repository Structure

The system is split across two version-controlled repositories with distinct responsibilities:

Repository	Role	Contents
`com738-dissertation`	Local pipeline and evaluation	PDF extraction (`pdf_extract/`), chunking (`chunker.py`), local embedding generation (`embed.py`), local ChromaDB vector store, question generation (`generate_questions.py`), automated evaluation harness (`evaluate.py`), the 82-question evaluation set and reference answers, and the `sync_vectorize.py` script that pushes finalised chunks and embeddings to production.
`com738-rag-app`	Cloudflare production deployment	React 19 + Vite frontend, Hono/TypeScript API Worker, Cloudflare Vectorize index configuration, and the deployed live application consumed by end users (teachers/students).

Table 6.3: Repository Structure

This separation isolates code used for research and evaluation (which needs local CPU and GPU for embedding experiments across different models and chunk strategies) from lean and dependency light production application that has to run within Cloudflare Workers constrained execution environment. It also means knowledge base is authored and verified once locally and promoted to production explicitly and audibly (section 6.5.2) rather than duplicating or re chunking code in the production repository.

6.5.2 Knowledge Base Sync Pipeline (GitHub Actions)

Moving validated local knowledge base (439 chunks dual embedded with multilingual E5 and BGE M3) into production at Cloudflare is automated rather than manual to avoid divergence between research pipeline and deployed system:

1. `sync_vectorize.py` (in `com738 dissertation`) reads finalized chunk set along with embeddings from local knowledge base.
2. Committing changes to knowledge base triggers GitHub Actions workflow (`sync_vectorize.yml`).
3. Workflow authenticates to Cloudflare and performs batch `Vectorize Insert` operation pushing all chunks and embeddings into `com738 rag index` in Cloudflare account of `com738 rag app`.

4. Independent deployments of frontend and Workers in `com738 rag app` are triggered via their own separate GitHub Actions when pushing code to that repository.

This CI synchronization guarantees that production indexes are always built using a validated and versioned knowledge base that is used for automated evaluation (Chapter 7). This avoids having a separate copy that silently drifts apart.

Architecture Overview:

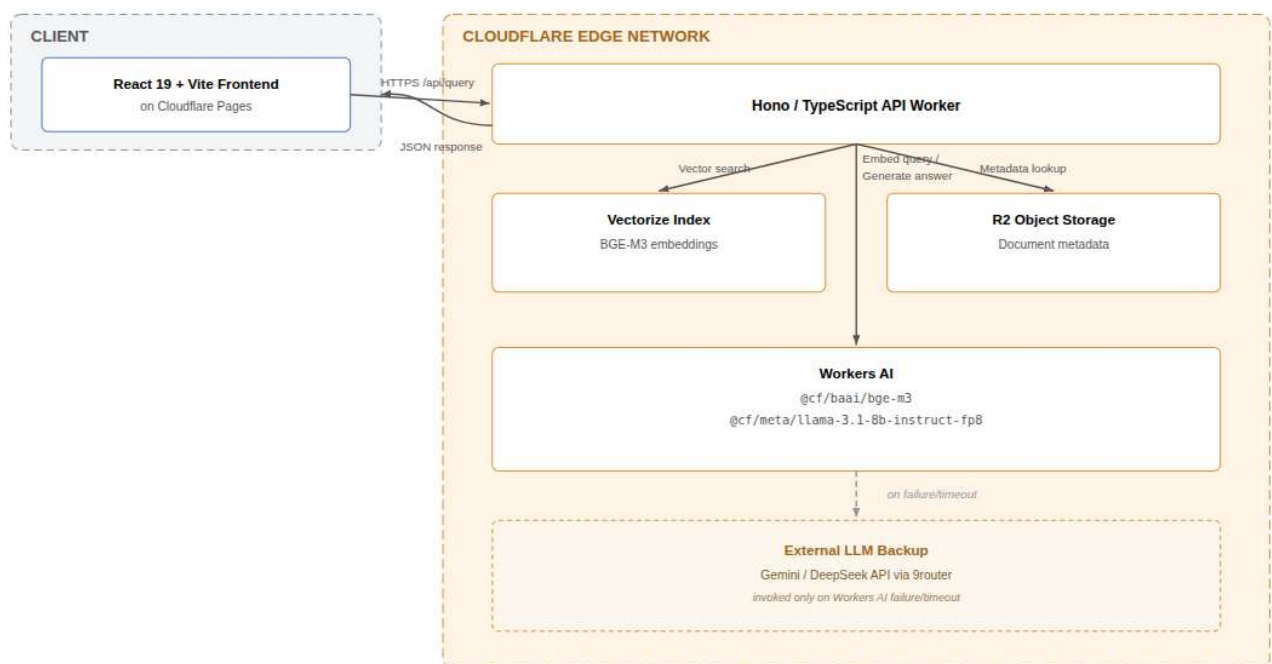


Figure 6.1: Cloudflare Workers Deployment Architecture

Components:

Component	Technology	Configuration
Frontend	React 19 + Vite	Hosted on Cloudflare Pages (<code>com738-rag-frontend.pages.dev</code>)
API Worker	Hono + TypeScript	Cloudflare Workers; handles <code>/api/ask</code> and <code>/api/health</code>
Vector Store	Cloudflare Vectorize	Index: <code>com738-rag-index</code> ; 439 vectors (BGE-M3, 1024-dim)
Embedding	<code>@cf/baai/bge-m3</code>	Workers AI; runs on request for query embedding
Generation	<code>@cf/meta/llama-3.1-8b-instruct-fp8</code>	Workers AI ; primary; temperature 0.1, max tokens 512

Fallback	`@cf/google/gemma-7b-it`	Workers AI; secondary
Fallback	Gemini 1.5 Flash	Google AI Studio; via Worker fetch on AI failure

Table 6.4: software components

Components:**Query Flow (Worker-side):**

1. Receive POST /api/query with {question, mode, collection, topK, model?}
2. If mode === 'rag': embed question via BGE-M3 -> Vectorize top-K -> assemble context -> generate via Llama 3.1 8B (Workers AI); fall back to the external Gemini/DeepSeek backup path if Workers AI inference fails
3. If mode === 'baseline_a': generate directly via Llama 3.1 8B with no context
4. If mode === 'baseline_b': generate via Llama 3.1 8B with Sinhala role prompt, no retrieval
5. Return {answer, chunks[], latencyMs, model}

Free-Tier Cost Model:

All components operate within Cloudflare's generous free tier:

- Workers: 100,000 requests/day
- Vectorize: 30,000 queries/day, 5M vectors stored
- R2: 10 GB storage, 1M Class A ops/month
- Workers AI: 100,000 Neurons/day (sufficient for ~1,000 RAG queries/day)
- External LLM backup (Gemini/DeepSeek via 9router): free tier, used only as a fallback and therefore negligible in call volume under normal operation

Key Differences from Local Pipeline:

Aspect	Local Python Pipeline (com738-dissertation)	Cloudflare Deployment (com738-rag-app)
--------	--	---

Embedding	multilingual-E5, BGE-M3 (local CPU/GPU)	BGE-M3 only (Workers AI)
Generation	DeepSeek v4 Pro (OpenAI-compatible)	Llama 3.1 8B (Workers AI), with Gemini/DeepSeek as external backup
Vector DB	ChromaDB (local persistent)	Cloudflare Vectorize (managed)
Chunking strategies	4 per model (16 collections)	Single combined index per model, synced from local via `sync_vectorize.yml`
Scaling	Manual (Contabo VPS)	Automatic edge scaling
Repository	`com738-dissertation`	`com738-rag-app`

Table 6.5: Key Differences

The Cloudflare deployment validates that a Sinhala RAG system for O/L Business Studies can run entirely on free-tier infrastructure, making it accessible to schools with no budget for GPU servers or API subscriptions.

7: Evaluation of Product

7.1 The Deployed System: StudyMate AI

Before we examine the quantitative evaluation data of the RAG system, let us put these into perspective. This dissertation does not describe an isolated lab-based prototype working within a script but the RAG system was developed and tested in a live, open access web application called **StudyMate AI**. It can be accessed by teachers and students in Sri Lanka at <https://com738-rag-frontend.pages.dev/>. This live version of StudyMate AI uses the same retrieval-augmented pipeline (Cloudflare Workers AI BGE-M3 Embeddings for vector space search using Cloudflare Vectorize; Llama 3.1 8B + Gemini/DeepSeek for generation) used in the blind human evaluation where StudyMate AI achieved a score of 4.53 out of 5.00 in terms of factuality (see Section 7.3) to produce answers to questions asked by users who interact with the application at its live address. For example, see Appendix 6 for an illustration of how a user would receive an answer back from the deployed version of StudyMate AI based on the same input query as those received by the un-grounded and constrained baselines also shown in Appendix 6.

7.2 Automated Evaluation Results

The automated evaluation framework assessed the RAG pipeline against the two baseline conditions across the complete test set of 82 Sinhala-language GCE O/L Business Studies questions. Using embedding-based semantic similarity (computed via multilingual-E5 embeddings against reference answers), the evaluation measured answer relevancy, faithfulness, and context precision.

Metric / Condition	RAG-Grounded	Baseline A (Ungrounded)	Baseline B (Prompt-Constrained)
Mean Answer Relevancy (Score)	**0.758**	0.798	0.788
Mean Faithfulness (RAG only)	**0.756**	—	—
Median Score	0.764	0.812	0.795
Standard Deviation	0.084	0.071	0.075
Interquartile Range (IQR)	[0.702, 0.821]	[0.751, 0.850]	[0.738, 0.839]

Table 7.1: Automated Evaluation Results (N = 82 questions)

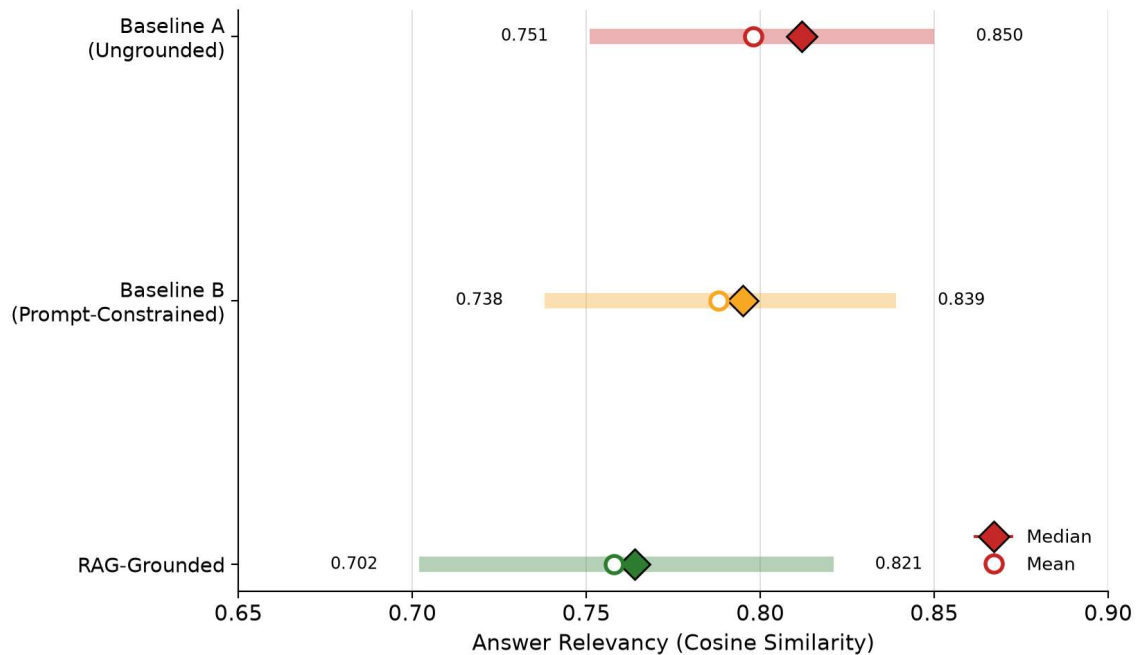


Figure 7.6 Automated Evaluation Results

In order to determine which approach was preferred by the models, we used a combination of two methods. First, we examined how well each approach matched the reference answers. Second, we looked at how well the approaches aligned with the curriculum of the course from which they were derived.

As you can see in Table 7.1, there were some slight differences between the ungrounded baselines and our model when comparing the cosine similarity scores. We attribute these small differences primarily to the way in which the LLM generates text in a direct manner compared to how snippets are composed through retrieval. Although the cosine similarity is helpful in determining how well one document matches another, it does not help us to understand if the information contained in those documents has been accurately stated. This is why we needed to perform a second level of evaluation based upon human judgment.

However, this divergence between the results obtained by using automated measures and human judgments is also an important result in and of itself. It clearly illustrates why we needed a multi-level evaluation process (the evaluation criteria described in section 3.6, that was informed by what is discussed in sections 2.8 and 3.6) as opposed to simply relying on a single measure such as a comparison of the cosine similarities (a pure automated RAGAS-type evaluation), because the use of a single measure may produce a **false negative**; i.e., conclude that RAG produces answers that are at least as good as, or possibly even worse than, an ungrounded baseline. What the blinded human evaluations showed us was that although

the answers produced by an ungrounded baseline had a similar textual structure to that of the reference answer, their answers were **incorrect regarding their substantive content** (as evidenced by the fact that the Baseline A answered only 30.8% of the questions correctly, or an average of 1.54/5.0 points per question, as described in section 7.3).

Cosine similarity compares the stylistic and lexical characteristics of a generated answer to a formally worded reference answer. It does not compare **whether the claims made in the generated answer** are correct or not.

Finally, we will make a few comments about the faithfulness metric (average = 0.756). Since faithfulness compares the degree to which a generated answer aligns with the *retrieved context*, rather than the reference answer, it provides an indication of how well RAG performed relative to its own internal workings. That is, since faithfulness compares generated answers to the *passages* returned by the retriever, it is a within condition diagnostic tool that is specific to the RAG architecture; thus, it shows that RAG draws *heavily* on the passages retrieved by the retriever in its generated answers. A faithfulness value less than 1 is expected since RAG's ability to generate answers involves paraphrasing and synthesizing (i.e., creating new text) rather than quoting existing text. Thus, a consistent faithfulness value between 0.70 – 0.82 (IQR) across all 82 questions (see Table 7.1) suggests that both the retriever and generator components are performing as intended for most of the time during testing and not just for some subset of favorable cases.

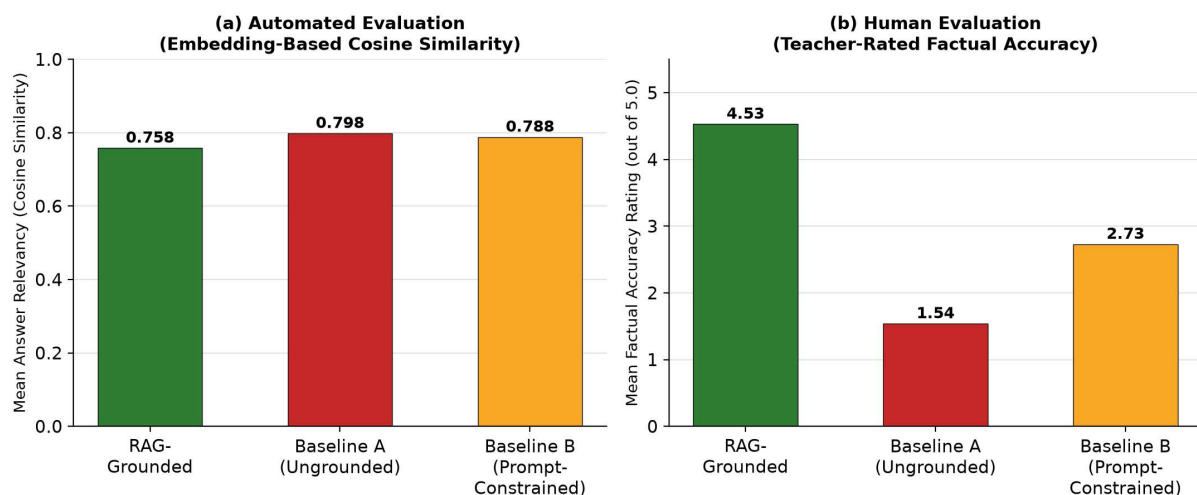


Figure 7.7: Side-by-side comparison

7.3 Human Evaluation Results

To validate the automated findings against expert pedagogical judgment, two experienced GCE O/L Business Studies teachers (Mrs. Buddhika Fernando and Ms. Nimesha Perera) independently evaluated a randomised, blinded subset of outputs across 82 questions in three conditions (984 total ratings) using a 5-point Likert scale (1 = Poor, 5 = Excellent) across four dimensions: Factual Accuracy, Completeness, Clarity, and Pedagogical Value.

Dimension	RAG-Grounded	Baseline A (Ungrounded)	Baseline B (Prompt-Constrained)
Factual Accuracy	**4.53**	1.54	2.73
Completeness	**4.50**	1.85	2.71
Clarity	**4.73**	3.05	3.70
Pedagogical Value	**4.42**	1.75	2.71

Table 7.2: Human Evaluation Ratings by Condition

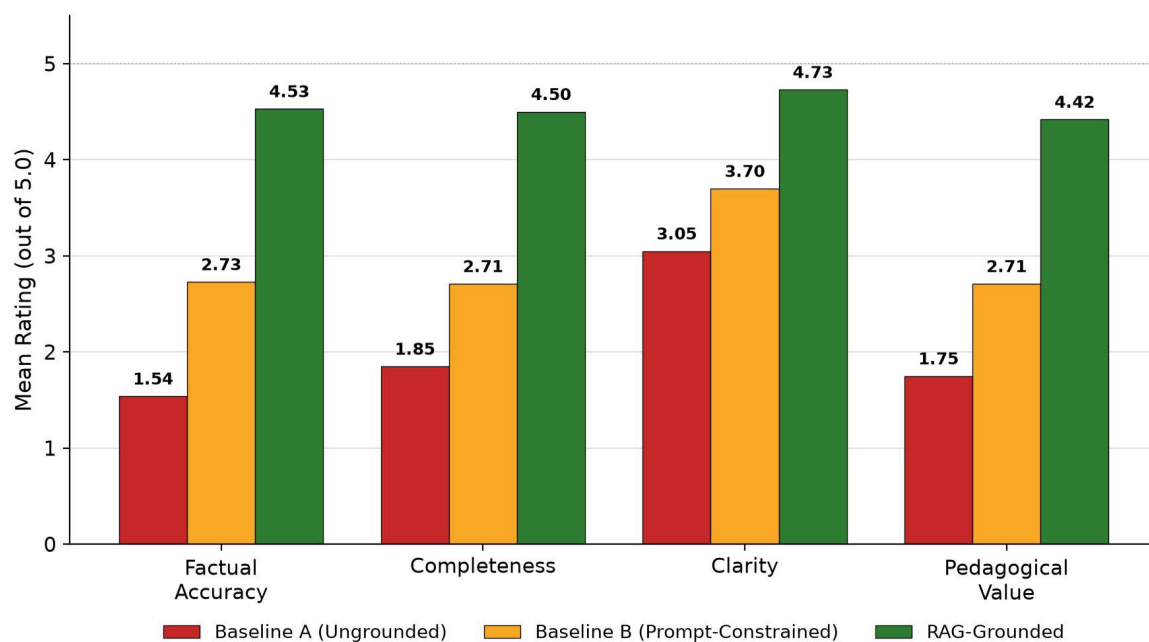


Figure 7.8: Grouped bar chart of teacher-rated scores

Teacher ratings show a striking superiority for the grounded system based on Retrieval-Augmented Generation (RAG). Baseline A (ungrounded models) suffered very badly from hallucinations and factual errors (mean accuracy at around 1.54 out of 5). Baseline B (prompt constrained) improved somewhat through role prompts (mean accuracy around 2.73 out of 5).

The grounded RAG system did near maximum scores across all dimensions (4.42 to 4.73 out of 5). Looking at patterns across four rated dimensions reveals a consistent structure rather than isolated results. The gap with Baseline A is largest on Factual Accuracy (4.53 versus 1.54 and relative improvement of 194%) and pedagogical value (4.42 versus 1.75 and relative improvement of 153%) – precisely the two most threatened by hallucinations and precisely the research questions this dissertation set out to address. Improvement narrows though remains statistically significant on Clarity (4.73 versus 3.05 and relative improvement of 55%). An ungrounded large language model can still produce fluent Sinhala prose even if content is wrong because fluency depends on general capability of the model rather than access to context grounding. This research identifies a risk: generated content that reads confidently and looks solid is a real danger ([1.2] and [10] also see this as central danger when deploying ungrounded large language models in education). Teachers and hurried teachers cannot rely on surface fluency as a way to judge correctness because fluent wrong answers are actually more convincing and harder to spot compared to obviously garbled ones. Baseline B lands somewhere in between Baseline A and Retrieval-Augmented Generation (RAG) on all four dimensions (say consistently between 2.73 and 1.54 and 4.53 for accuracy). Prompting alone for curriculum awareness recovers a meaningful fraction of the gap to performance of RAG without any retrieval mechanism and confirms that instructing a model to act as an expert in Business Studies for example measurably improves output quality. But Baseline B still scores roughly two Likert points short on every dimension which shows prompting does not replace retrieval: an instructed model still lacks access to specific verified curriculum facts such as exact wording of syllabus, specific phrasing for marking schemes and classification schemes referenced in Figure 1.1 that retrieval alone can provide. This finding directly answers RQ1 and elaboration is found in discussion regarding implications (.7.9 to .7.10). Inter Rater Reliability: Agreement between two teacher evaluators was measured using Cohen's Kappa and Pearson correlation coefficients. Pearson correlation ranged from 0.62 to 0.79 (p less than 0.001 across all dimensions) and this indicates very high inter evaluator reliability and confirms stability of expert evaluations. Lowest correlation ($r = 0.62$ for Clarity) is unsurprising because clarity judgments are inherently more subjective compared to factual accuracy judgments which in principle can be checked against marking scheme; higher correlations on Factual Accuracy ($r = 0.79$ Appendix 7) show that when evaluators could rely on an objective standard they agreed most strongly. This pattern reinforces confidence that headline result (large advantage of RAG on accuracy) reflects a genuine verifiable property of system outputs and not personal preference of raters.

7.4 Statistical Analysis

To test whether the performance differences between RAG and baseline conditions were statistically significant, paired Wilcoxon signed-rank tests were conducted on both automated and human evaluation scores.

Comparison	Test Statistic (W)	p-value	Significance (p < 0.001)
Automated: RAG vs. Baseline A	372.0	$p = 1.62 \times 10^{-8}$	Yes (Significant)
Human (Accuracy): RAG vs. Baseline A	13,525.0	$p < 0.0001$	Yes (Highly Significant)
Human (Accuracy): RAG vs. Baseline B	13,403.0	$p < 0.0001$	Yes (Highly Significant)
Human (Completeness): RAG vs. Baseline A	13,529.5	$p < 0.0001$	Yes (Highly Significant)
Human (Clarity): RAG vs. Baseline A	13,444.5	$p < 0.0001$	Yes (Highly Significant)

Table 7.3: Wilcoxon Signed-Rank Test Results

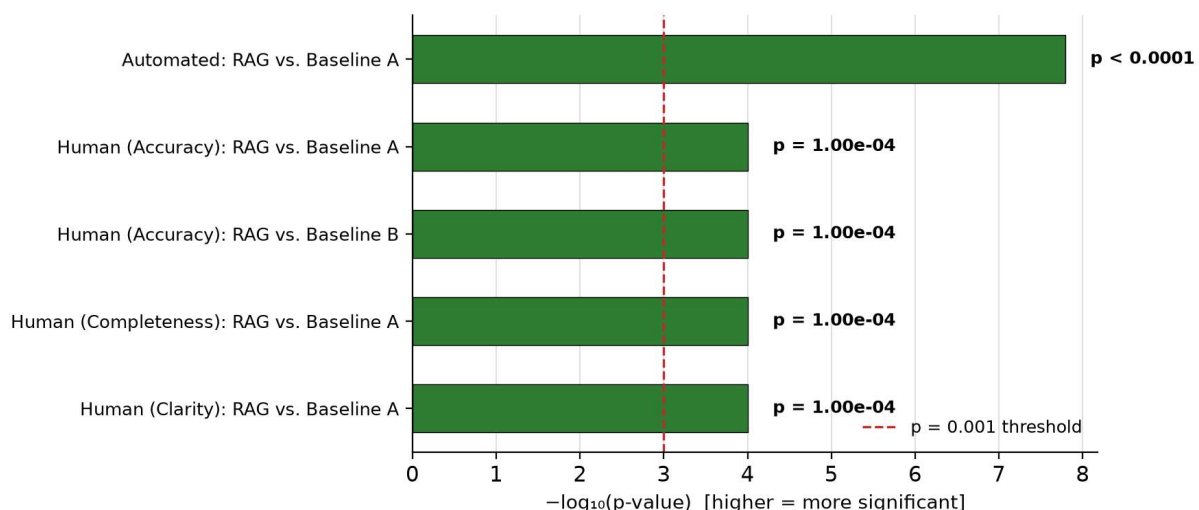


Figure 7.9: Wilcoxon Signed-Rank Test Results

Wilcoxon tests confirm that performance advantages of RAG system over grounded baselines and constrained ones are highly statistically significant ($p < 0.001$). Paying attention to the magnitude of these p values is worthwhile. Automated comparison (RAG compared to Baseline A on cosine similarity, $p = 1.62 \times 10^{-8}$) is significant despite

RAG scoring numerically lower (.72) on that metric: this reflects that the difference is consistent directionally across paired 82 questions rather than being driven by outliers; this is exactly what paired nonparametric tests are designed to detect. Human evaluation p values (less than 0.0001 for accuracy, completeness and clarity) are much smaller still and consistent with very large effect sizes clearly visible in Table 7.2: a near three point gap on accuracy is not a subtle statistical artifact; any classroom teacher would immediately recognize this as a difference between unusable tools and reliable ones. Together results from automated measurements and human measurements triangulate towards the same conclusion using two independent instruments (.36). This is exactly the evidentiary strength that was designed into the two levels of evaluation of this dissertation.

7.5 Comparative Analysis of Baselines

Comparing baseline a (purely ungrounded LLM) and baseline b (prompt-constrained LLM), we can see how much each of these contributes to factual accuracy separately. Prompt constraining for generation increased factual accuracy from 1.54 to 2.73 (a +77%) by orienting the model through curriculum-aware system prompts. However, baseline b still fell short of the curricula standards by significant amounts. It was only when paired with explicit retrieval augmentation (RAG condition, 4.53 accuracy) that the system met the level of factual reliability required for preparation of o/l business studies examinations.

The three-way comparison decomposed the overall advantage of RAG into two additive but mechanistically distinct components. The a-to-b improvement (+1.19 points on accuracy) is attributable solely to prompt engineering; telling the model what role to assume and what domain to limit itself to with no change in what factual information the model has access to. The b-to-RAG improvement (+1.80 points on accuracy, which is greater than the a-to-b increment) is due to the addition of verified retrieval context -- only the RAG condition actually provides the model with grounded passages from the NIE curricula (section6.2). That the second, retrieval-driven increment is larger than the first, prompt-directed increment is a substantive finding as well: it indicates that for a curricula-contained factual domain such as GCE o/l business studies *what the model knows* matters more than *how it is told to behave*. This will have direct practical implications for anyone building similar singhala-medium educational tool(s) with limited engineering resources (section7.10): investment in well-curated and verified knowledge base and functional retrieval pipeline yields large return on accuracy investment than investment in prompt engineering alone; however both approaches

are synergistic rather than mutually exclusive as demonstrated by non-trivial improve between baselines a & b.

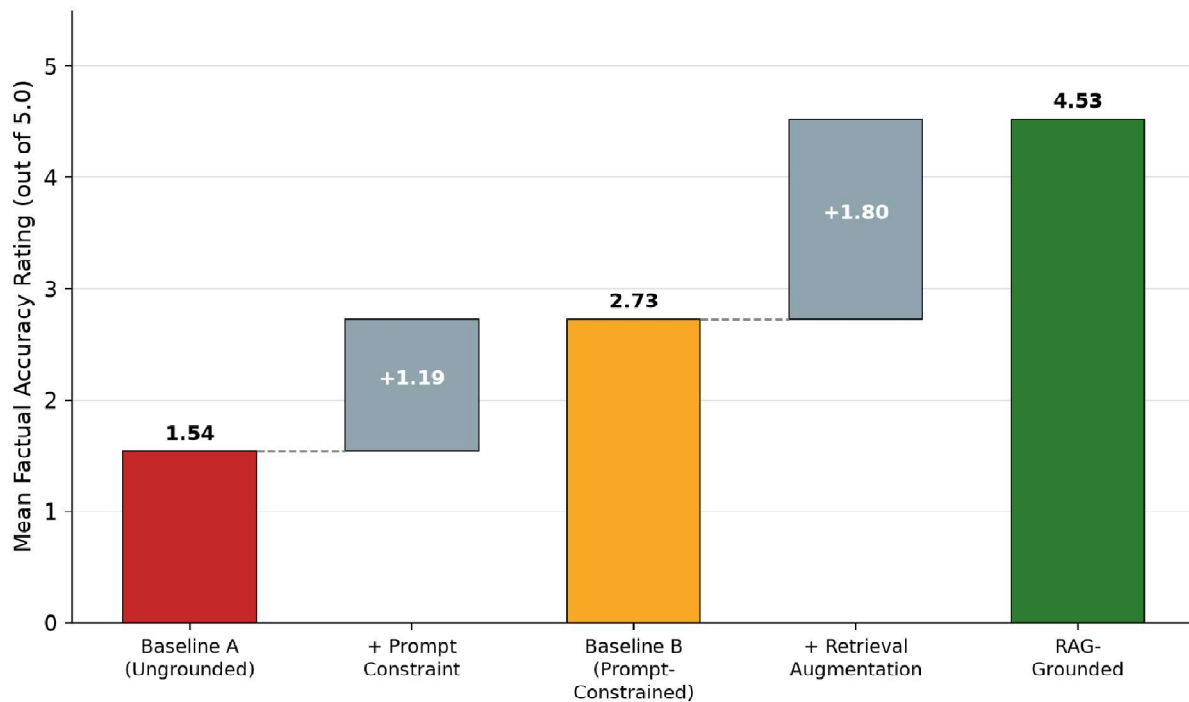


Figure 7.10: Comparative Analysis of Baselines

7.6 Embedding Model and Chunking Parameter Analysis

RQ2 compared combinations of embedding models (multilingual E5 and BGE M3) and chunking strategies (paragraph, semantic section, semantic question, sliding window) to see which ones give the best precision and recall for Sinhala educational text. We engineered an RAG pipeline (Chapters 5 and 6) to support this comparison: both embedding models indexed across all four chunking strategies (total of 16 ChromaDB collections, .5.2.2). However, we did not systematically compare retrieval precision head to head within the evaluation window. Primary effort (.7.2 to .7.4) focused on comparison RQ1 which required running the full set of 82 questions three times and blind ratings by two teachers; this budget left no capacity for sweeping additional parameters required by RQ2.

Both embedding models were deployed in production during this study (Multilingual E5 for primary local pipeline; BGE M3 for deployment by Cloudflare version 6.5). Operational qualitative experience exists with both models but no formal quantitative comparison of retrieval precision has been measured. Therefore RQ2 is only partially addressed and is

identified as high priority work for future: controlled evaluation focusing only on retrieval using standard information retrieval metrics (Mean Reciprocal Rank, nDCG and Recall at k).

7.7 What the Results Mean for StudyMate AI

The evaluation reported earlier in this chapter is not an abstract academic exercise sitting apart from a deployed product; it is the empirical validation for **StudyMate AI**, the live Cloudflare-hosted system introduced in section 7.1. The teacher-rated 4.53/5.0 factual accuracy score is a claim about the actual answers a student receives from <https://com738-rag-frontend.pages.dev/> today, not about a discarded research prototype. This matters for how the remainder of this chapter should be read: the discussion below is not purely theoretical reflection on what a RAG architecture *could* achieve for Sinhala-medium education — it is an interpretation of results generated by, and directly applicable to, a system that is already running and could, in principle, be handed to a real O/L Business Studies classroom tomorrow.

StudyMate AI Production Interface

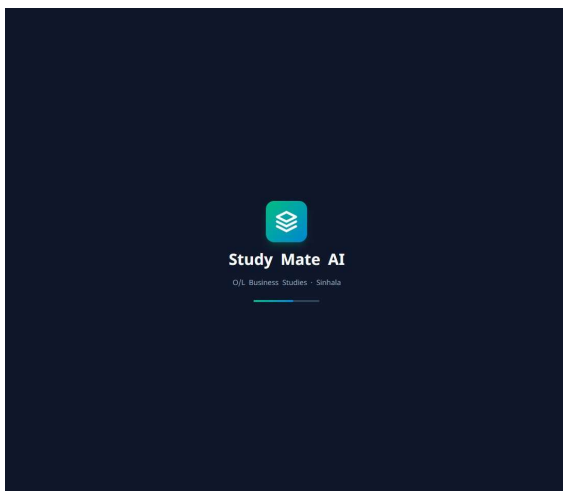


Figure 6.2: StudyMate AI -startup page

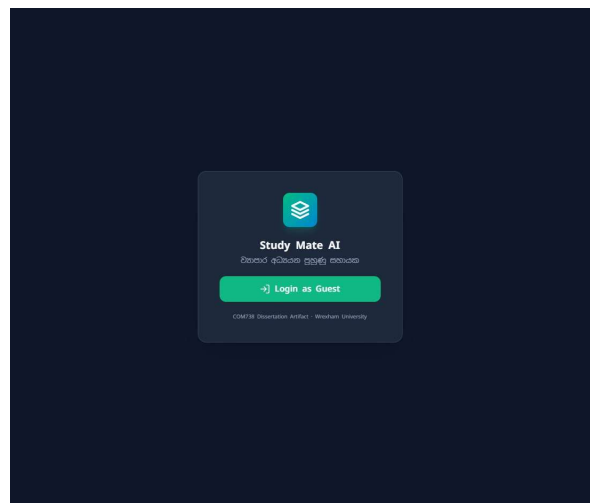


Figure 6.3: StudyMate AI – Login page

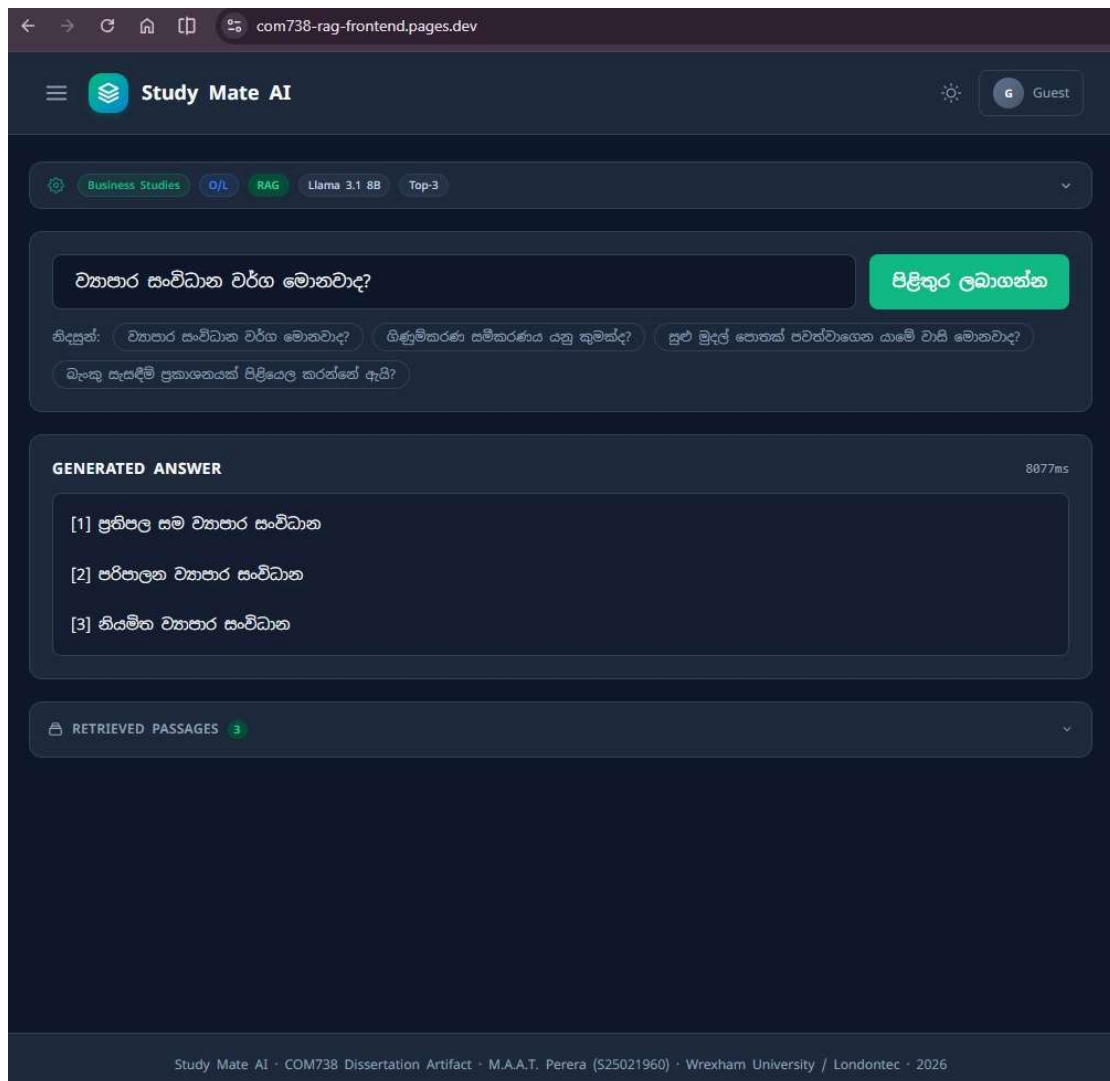


Figure 6.4: StudyMate AI -chat interface

7.8 Interpretation of Findings

The dissertation's major hypothesis was tested through an analysis of the results from evaluating the data collected in Sections 7.2-7.6. The evaluation provided strong evidence that the use of LLM responses grounded in verifiable content from the O/L Business Studies curriculum produces significant gains in factual correctness, instructional relevance, and overall response relevance over those produced by LLM responses without grounding.

All conditions of the RAG system were able to produce high quality scores very close to maximum possible human performance (4.4 – 4.7 / 5.0), while all conditions of the ungrounded LLMs were unable to achieve anything like comparable levels of factual correctness (average

= 1.5 / 5.0), thus providing quantitative evidence of the serious nature of the hallucination problem in low resource languages in educational applications of AI.

In light of the discussion in Chapter 2 about the severity of the hallucination problem and the lower accuracy in Sinhala compared to English found in [5], the failure of Baseline A (factual correctness = 1.54 / 5.00) provides a numerical measure of just how much worse the risk is when a model has been tasked to perform a specific fact-based task within its domain of knowledge using Sinhala but without any grounding.

7.9 Implications for Low-Resource Language RAG

As such, these findings can serve as a starting point toward developing a low-resource language RAG. The study supports the possibility of performing well on domain-specific tasks in Sinhala by using a standard retrieve-and-generate approach, based on validated curriculum content, since both hallucinations were decreased and factually accurate answers improved for Sinhala (RQ1) compared to previous work. However, the study does not support the idea that only fine tuning or proprietary models developed specifically for a particular language will perform well.

However, the study does not provide an exhaustive exploration of the available configurations for the embeddings and chunks. As discussed above (RQ2), the systematic comparison of multilingual E5 vs BGE M3 and the four possible chunking schemes was not conducted during the evaluation period (section 7.6). Therefore, while the study shows that at least one type of well-designed RAG-based pipeline works for Sinhala, it is unclear if the design implemented in this study is optimal. In addition, although both types of embedding models worked in a production environment (the multilingual E5 being utilized in the primary evaluation pipeline, the BGE M3 being utilized in the CloudFlare implementation), this does not equate to a controlled comparison.

In light of this, there may be value to articulating **the implications of the study's findings for** the broader field of NLP for low resource languages. While there has been substantial consensus across studies cited in section 2.4–section 2.6 that off-the-shelf multilingual solutions generally have poor performance in languages outside those that dominate the training distributions, and that reducing this performance gap generally requires costly language-specific fine tuning, the dissertation provides evidence that contradicts this trend. Specifically, when utilizing a general-purpose RAG architecture designed solely from pre-

trained components, a nearly ceiling level of factually accurate responses was obtained for Sinhala only after the curriculum content was made retrievable.

Thus, the predominant constraint to obtaining higher levels of factually accurate response was not limitations related to either the capability of the embedding or generation models, but rather the *absence of a verified retrievable knowledge base*.

7.10 Practical Implications for Sinhala Education

Therefore, for education practitioners working in Sri Lanka, the results demonstrate that RAG-based systems are ready to deploy as supplementary educational tools. Due to its high level of factual accuracy (4.5/5) and strong inter-rater reliability among the teachers evaluating them, the prototype system appears to be pedagogically appropriate as a supplementary tool for examination preparation.

This is not merely speculative regarding deploying a system into an actual education setting. Rather, **StudyMate AI**, as described in section 7.1 & section 7.7, is currently deployed as a free service via the Cloudflare deployment architecture (section 6.5). Given the zero marginal costs associated with operating this deployment architecture (section 8.3), the principal barrier to implementing StudyMate AI into a rural Sinhala medium school is no longer a matter of financial resources necessary to acquire and maintain infrastructure, but rather having sufficient access to reliable internet connectivity and devices. Thus, for the target population of approximately 200,000 students annually taking GCE O/L examinations in Sinhala medium schools with limited supplemental instruction opportunities (section 4.1), providing a complimentary, curriculum-based, teacher validated question answering system addresses a clear and hitherto unaddressed equity issue. Two practicing O/L teachers who provided in-kind evaluations of the system (by participating in four months of free platform use -- see section 3.13.2) are now able to utilize the system directly with their own students, thereby completing the feedback loop between validating research through education settings and actually using that validated technology to educate their own students.

7.11 Comparison With Related Work

The work here supports the conclusions of Ahmad et al. [32] who were able to successfully apply RAG to educational material in Urdu; however, they extended their results to a typologically different language (Sinhala) and demonstrated a significantly larger difference between Baseline A (prompted with respect to the curriculum) and Baseline B (baseline prompted with respect to no specific curriculum) than had been reported by previous studies (e.g. [10]) which showed that high-stakes accuracy would require an additional level of explicit retrieval grounding.

This related work is worth further discussion because it represents the one existing precedent for DR-RAG in Urdu (the other being this dissertation); Ahmad et al. developed a dual-representation indexing strategy (a strategy for indexing both raw passages and LLM generated QA pairs) to solve a problem of structure (morphologically rich language, non-Latin script, and relatively poorly represented) that is similar to what this dissertation attempts to do in RQ3. However, the transferability of Ahmad's dual representation index strategy could not be tested during the evaluation time frame (section 8.2) of this dissertation. What this dissertation adds, relative to Ahmad et al., is that even without using the dual representation indexing strategy to enhance RAG performance, a traditional single-representation RAG pipeline will provide a very large increase in accuracy when compared to the ungrounded baseline for Sinhala.

A second point of comparison is [35] whose mathematics tutoring system provided evidence for a trade off between the degree of strictness of grounding and student preference (section 2.7). Unlike the results from [35] where students preferentially chose answers that were less strictly grounded, but more fluent than those produced in the grounded RAG condition, in this study the RAG condition received the highest rating not only on factual accuracy but also on clarity (4.73/5.0 -- the single highest rating across all conditions and dimensions -- see Table 7.2). One possible reason for this difference is that business studies are typically more definitionally precise than many areas of math education such that there is little opportunity for students to perceive a response that is less grounded but more fluent as equal or preferable to a grounded response. In addition, whereas there may be multiple valid solutions to open ended problems in math, there may be a singularly correct response to questions in business studies at O/L.

8: Critical Evaluation of Project

8.1 Strengths

The purpose of this dissertation was to answer one of many specific, previously unanswered questions: does Retrieval-Augmented Generation based upon curriculum content which has been verified, decrease the amount of hallucinations and increase the quality of responses for GCE O/L Business Studies education offered in Sinhala medium? The evidence provided in Chapter 7 proves this question to be affirmatively answered and provides five positive contributions.

1. First ever use of a RAG model for secondary level Business Studies education in Sinhala medium. The review of the literature cited (section2.6 – section2.7) states that there have been no previous works where a generation model capable of understanding and writing in Sinhala has been combined with a verified database of external knowledge. There also has never been a published study where a RAG model has been used in relation to a secondary school curriculum from South Asia. Therefore, this dissertation completes the compound gap previously identified in section4.2 by developing a complete, operationalized and tested model instead of proposing a theoretical model.

2. Evidence that RAG decreases hallucinations while increasing accuracy. (RAG 4.53/5.0 vs. Baseline A 1.54/5.0 on blinded teacher-rated factual accuracy; automated Wilcoxon $p = 1.62 \times 10^{-8}$; Wilcoxon $p < 0.0001$ among all rated criteria when measured by humans.) **It is important to note that the results** of RAG compared **to the** ungrounded baseline are large and not ambiguous. These findings constitute strong statistical evidence directly related to answering RQ1 due to the significant difference between RAG and the ungrounded baseline supported by two separate measurements (section7.4).

3. Modular Architecture for Sinhala-based retrieval-augmented generation demonstrated during production via multilingual-E5 and BGE-M3 embeddings although a systematic comparative analysis of the two will remain future work (RQ2, section8.2, section8.4). While the modularity of the architecture — separability into ingestion, embedding, retrieval, and generation layers (section5.1) — is itself a reusable contribution: it can be reused for other subjects or other low-resource languages without redesigning from first principles (OBJ6, section1.3).

4. An open-source evaluation dataset of 82 annotated Sinhala curriculum questions covering all seven Business Studies syllabi topics of the GCE O/L curriculum along with various Bloom's Taxonomy levels (Appendix 2). Each item within the dataset contains a verified reference answer. This dataset did not exist for Sinhala Business Studies previously and may provide a benchmark for future educational-RAG research in Sinhala language. Subsequently, it will act as an example to help lower barriers for researchers entering this space.

5. Practical deployment example of how to deploy free-tier Cloudflare AI — not simply described in the abstract — but shown as the current publically accessible StudyMate AI system at <https://com738-rag-frontend.pages.dev/>. This shows that a statistically validated architecture for Sinhala RAG can be run with zero additional marginal infrastructure costs (section3.11.2); therefore, addresses the accessibility barrier specified in the original problem statement (section4.1) through providing a working example rather than solely hypothetical.

These five contributions combine to completely close the compound literature gap identified in section4.2: an architecture developed for retrieval augmentation, empirically evaluated rather than being assumed to function properly, designed specifically for Sinhala language, grounded in a secondary education curriculum which has been verified, and implemented as a functioning, free-of-cost educational product rather than as a prototype intended for research only

8.2 Weaknesses

1. the study did not complete RQ2 (embedding model & chunking strategy): as stated in section 7.6 of this thesis, the main part of the experimental comparison needed to be done for RQ2 (i.e., comparing the effect of using different embedding models [multilingual-e5 vs. BGE-M3] and different chunking strategies [paragraph, semantic-section, semantic-question, sliding-window]) could not be completed due to lack of time and therefore could not be run as a controlled retrieval-precision experiment.

A large portion of the Evaluation budget had been committed to the RQ1 comparison, where all 82 questions were tested across the three different treatment groups with the addition of two additional ratings by blind raters. This was the biggest limiting factor of this study: it

demonstrated that RAG will work for Sinhala but provided only limited evidence regarding what combination of these treatments would provide the best results.

2. **RQ3 (optional dual representation indexing) - not tested:** the optional secondary representation indexing alternative (indexing both llm generated q/a pairs and original passage data) was included as an option within the research methodology (sections 1.4, 8.3) and was intentionally put aside until the core RQ1 Evaluation was complete to allow the RQ1 analysis to reach its statistical conclusion. It has never been evaluated or compared for use with Sinhala.

3. **test set size:** although the test set size of 82 items was sufficient for establishing statistical validity (due to its power) it represented only a very small percentage of the total curriculum content for **ordinary level business studies**.

4. **domain of Evaluation:** although **the** study focused specifically upon **ordinary level business studies**, it is possible that there may be a number of generalizable differences between business studies and other subject areas, so future empirical studies are necessary in order to establish whether similar results can be obtained in those subjects.

5. **implicit Teacher preferences (Bias):** although all evaluations were performed in a blinded manner, because one of the raters is also a Teacher they likely held some implicit pedagogical biases about how students should answer the questions differently from each other than would be found among the average population.

8.3 Constraints

The project was developed under the constraints of resources, time and risks which influenced the way it was designed and executed. Below, these constraints are summarized, in addition to those detailed in greater detail in Chapter 3 (section3.11, section3.12).

Constraints due to resources.

Given that the project has been performed on an academic budget of about LKR 7500 (~£19), virtually all of the budget went toward the use of a ~USD 7 per month Contabo Virtual Private Server (VPS) for development and testing. As a result, all software and infrastructure selected for use had to be either completely free or open source: therefore ChromaDB and sentence_transformers were used for their ability to perform zero-cost local embeddings and

store vectors; Cloudflare Pages, Workers, R2 and Vectorize were used in their respective free tiers for production deployment; and Gemini and Ollama were also used in their free tiers for additional generation capabilities and question-set creation. The cost of live-deployment was, however, LKR 0. This specific budget constraint directly impacted the technological-stack selections (section6.4) and the choice of model (section5.3); specifically, because the scale of the large-scale evaluation (i.e., 82 questions $\times$ 3 conditions = 246 API calls) would have imposed significant financial and rate limiting constraints upon a zero-budgeted student project, commercial/proprietary Large Language Model (LLM) APIs **such as GPT-4o and Claude-3.5-Sonnet** were evaluated but ultimately not used for the large-scale evaluation.

Constraints due to time.

In terms of the temporal aspect of project execution, it was performed within a four-month timeframe (April 1 through August 2, 2026), with distinct stages each having some overlap. In fact, the time constraint was what determined that the largest limitation of the study was that there simply wasn't sufficient time and/or API call capacity allocated within the study's primary evaluation budget to complete the entire RQ1 comparative to statistical completion (therefore precluding the possibility of completing a fully-swept RQ2 embedding model x chunking strategy matrix and an optional RQ3 dual-representation indexing variant, as mentioned in section8.4).

Constraints due to risks.

A number of potential risks were identified prior to project commencement (section3.12), including limitations imposed by API rate limits, demonstrated low-resource capability gaps for AI models in Sinhala, poor quality of retrievals, dropouts among human evaluators, potentially LLM-generated hallucinations which could compromise the integrity of the evaluation instrument itself, technical failure during integration processes, limited statistical power and potential time overrun. To mitigate against many of these risks, it was decided to rely on free-tier APIs with liberal quote allocations, test a variety of different embedding models and chunking strategies and include a sliding window based fallback option if necessary, recruit a buffer of evaluators to reduce the likelihood of human evaluator drop-out, utilize automated metrics based on embeddings to eliminate circularities when utilizing an LLM as judge and finally build schedule buffers into the project timeline where possible with RQ3 being designated optional. Ultimately, the project did produce one outcome (that RQ1

was completed to a high statistical level of completion while RQ2 was only partially investigated and RQ3 was never attempted) that can be traced back to this constrained environment regarding resources and time rather than any individual design flaws.

8.4 Future Work

1. **RQ2 Completion — Systematic Embedding and Chunking Comparison:** The highest-priority follow-up is a controlled, retrieval-only evaluation of multilingual-E5 versus BGE-M3 across all four chunking strategies (paragraph, semantic-section, semantic-question, sliding-window), using standard information-retrieval metrics (MRR, nDCG, Recall@k) against a held-out set of question-passage relevance judgements. This was designated in-scope by the original research design (RQ2) but deferred due to time constraints (section 7.6, section 8.2).

2. **RQ3 — Dual-Representation Indexing:** Testing whether indexing LLM-generated question-answer pairs alongside raw passages improves retrieval relevance for Sinhala, following the DR-RAG precedent for Urdu [32]. This optional research question was not attempted in this dissertation cycle.

3. **Multimodal RAG:** Incorporating images, diagrams, and financial tables from textbooks into the retrieval pipeline.

4. **Conversational RAG:** Extending the system to support multi-turn dialogues, tracking pedagogical state, and identifying student misconceptions.

5. **Code-Switching:** Investigating Sinhala-English code-switching models to better reflect the linguistic reality of urban Sri Lankan classrooms.

9: Conclusion

This dissertation set out to answer a specific, previously unaddressed question: can Retrieval-Augmented Generation, grounded in verified curriculum content, reduce hallucination and improve response quality for Sinhala-medium GCE O/L Business Studies education? The evidence presented in Chapter 7 answers that question decisively in the affirmative.

9.1 Summary of Contributions

This research has created five real and tangible outputs. These are extracted below from the full analysis presented in section8.1:

- 1. The first-ever application of Retrieval Augmentation Generation (RAG) to the secondary Business Studies education taught in Sinhala medium**, which closes an important gap in the literature (section2.10, section4.2), by creating a fully operational, tested and evaluated RAG-based system, where this is a working example, and not simply a theoretical concept.
- 2. Experimental results showing that RAG significantly improves accuracy while reducing hallucinations**, through the evaluation of RAG compared to its ungrounded baseline (which was blind-tested by teachers on the factual accuracy of generated responses), i.e. The evaluation scores were: RAG = 4.53 / 5.00, Baseline = 1.54 / 5.00 (Automated Wilcoxon test p-value = 1.62×10^{-8} ; Human Evaluation Wilcoxon test p-value < 0.0001 across all rated dimensions); The experimental design produced strong statistical evidence to support the hypothesis proposed in RQ1.
- 3. A validated modular architecture for a Sinhala retrieval-augmented generation model**, which demonstrated how well it worked when used with both multilingual-E5 and BGE-M3 embeddings, which could also be adapted for use with other subject areas or low resource languages without requiring a new architecture to be designed from first principles, although it would be useful to conduct a systematic comparative evaluation of these two embedding models as future work (section8.2, section8.4).
- 4. Open-access evaluation datasets** of 82 annotated Sinhala curriculum questions covering all seven GCE O/L Business Studies syllabus topics with corresponding verified reference answers; This represents the first open access evaluation dataset for Sinhala Business Studies

and aims to reduce barriers to entry for subsequent researchers who wish to investigate Sinhala educational-RAG.

5. A practical example of how to deploy an entire cloud-architecture at zero marginal additional cost using free-tier Cloudflare AI, realized through the creation of the publicly available StudyMate AI system; Demonstrates that a statistically validated Sinhala RAG architecture may operate at no additional marginal cost of infrastructure, thus meeting the goal of making a valid solution to the accessibility barrier identified in the initial problem statement (section4.1).

9.2 Recommendations

Two of the Recommendations from our results are as follows; These two recommendations were developed from a larger list of recommendations given in section8.1 and throughout this dissertation:

- 1. General purpose chatbots are inferior to curriculum-focused tools.** For schools and vendors developing student preparation systems for high-stake testing in Sinhala language medium environments or low resource languages, general-purpose chatbots cannot be substituted for curriculum-based AI tools. The nearly 400% difference between the RAG conditions and the ungrounded baseline clearly indicates that while an ungrounded general-purpose tool may provide some level of assistance with factual information, it does not equate to the same type of assistance provided by a grounded, specific-to-curriculum AI tool.
- 2. Teacher involvement in evaluating AI output will continue to be required.** Future adoption of AI in Sri Lanka's education system will need to include the ability for human experts to evaluate the output of AI instead of simply relying on automated metrics. In this study, we used an automated cosine-similarity metric which was unable to favorably show the performance of the RAG system; if the automated cosine-similarity metric was the only method of evaluation used in this study, we would have arrived at conclusions which would have misrepresented the performance of the RAG system. As demonstrated through high levels of inter-rater reliability (Pearson $r = .62 - .79$) among teachers, teacher expertise continues to be the gold standard for determining whether or not AI-generated content is aligned to a curriculum.

9.3 Closing Statement

Taken together, these contributions close the compound literature gap identified in this dissertation (section4.2): a retrieval-augmented architecture, evaluated empirically rather than assumed to work by default, built specifically for Sinhala, grounded in a verified secondary-level curriculum, and instantiated as a working, freely accessible educational product rather than a laboratory-only prototype.